

# All or Nothing? The Effects of Small-Scale Early College on Educational Attainment<sup>\*</sup>

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## Abstract

The full-school early college model has been shown to increase students' educational attainment. I investigate whether an early college program that enrolls a small share of students at conventional high schools can have a similar impact. Using student-level administrative data on K-12 and post-secondary course taking, I analyze the impacts of Early Middle College programs on educational attainment in Michigan. I instrument for EMC enrollment with 9th-grade cohort-level EMC eligibility at each high school. EMC eligibility increases the likelihood that students earn associate degrees. This is driven by EMC students taking more credits at 2-year institutions earlier in their educational careers. Analysis of mutually exclusive pathways suggests that EMC enrollment shifts students from 4-year-only enrollment to 2-year-only enrollment.

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<sup>\*</sup>This research result used data structured and maintained by the MERI-Michigan Education Data Center (MEDC). MEDC data is modified for analysis purposes using rules governed by MEDC and are not identical to the data collected and maintained by the Michigan Department of Education (MDE) and/or Michigan's Center for Educational Performance and Information (CEPI). Results, information and opinions solely represent the analysis, information and opinions of the author and are not endorsed by, or reflect the views or positions of, grantors, MDE and CEPI or any employee thereof. The research reported here was supported by the Institute of Education Sciences, U.S. Department of Education, through Grant R305B200009 to Michigan State University. The opinions expressed are those of the author and do not represent the views of the Institute or the U.S. Department of Education.

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# 1 Introduction

In recent years, programs designed to allow students to dual enroll in college courses during high school have proliferated across the United States. Early college provides one particularly intensive program of post-secondary study during high school. In the standard early college model, a high school (often located on a college campus) partners with a local institution of higher education (IHE) to offer students the opportunity to take a large number of college courses within their high school schedule at no cost to the students. Because they allow students to undertake their initial post-secondary enrollment within their high school’s institutional support structure, early colleges are designed to make college more accessible to students who otherwise might not have persisted beyond high school.

Several lottery-based studies have shown that the early college model can substantially boost degree attainment among program participants. (Haxton et al. 2016; Edmunds et al. 2020; Song et al. 2024) These studies track students who are admitted to high schools located on college campuses, designed to have 100 percent of their students participate in early college.<sup>1</sup> Atchison et al. (2021) uses the results of admissions lotteries to judge the cost-effectiveness of whole-school early colleges, finding that the lifelong earnings benefits of enrolling in early college outweigh the increased cost of providing early college in place of standard high school by a ratio of at least 4.6 to 1. As evidenced by the need for lotteries, these programs are oversubscribed, suggesting very high local demand. A second set of studies uses identification strategies based on matching observable characteristics between program participants and non-participants, similarly finding positive effects without requiring excess demand. (Fuller, Lauen, and Unlu 2023; Lauen et al. 2017; Swiderski 2024; Unlu et al. 2021)

Despite this strong record of effectiveness, districts face several barriers to starting early colleges. Most notably, creating a new school on a college campus requires considerable

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<sup>1</sup>Some early colleges are also more strongly focused on vocational training. Rosen et al. (2020) analyze P-TECH 9-14 programs in NYC that partner with both 2-year college programs and employers to incorporate work-based learning and college courses over a 6-year period.

resources, proximity to an IHE, and an active partnership with the participating IHE. In Michigan, where only 56 percent of districts fall within a community college district, many districts do not have these connections or proximity to an IHE. Additionally, students in districts with no in-district community college do not have access to preferred tuition rates at any two-year IHE.<sup>2</sup>

Early Middle College programs (EMCs) in Michigan provide one example of an early college design that creates partnerships between high schools and IHEs to allow students to embark on an intensive dual enrollment pathway without requiring substantial changes in school infrastructure or large commitments from participating IHEs. EMC students commit in 10th grade to enroll in an increasing number of existing community college courses as they progress through 11th and 12th grades, culminating with an associate degree and high school diploma awarded simultaneously after a fifth year of high school. EMCs generally serve a small portion of the student population at participating high schools, so they require minimal adjustments to staff, organization, and school culture. They partner with community colleges to offer EMC students access to a wide range of existing courses on the college campus or online.<sup>3</sup> Therefore, EMCs require relatively little investment beyond the local administrative burden of complying with state reporting requirements and securing an IHE partner.

EMCs do not seem to be limited to districts with close IHE proximity. In fact, there is at least one EMC in 49 percent of districts that fall outside any community college district, compared to 54 percent of districts within a community college district. This suggests that the EMC model is attainable for districts with and without existing links to a particular college. So, can a form of early college that uses existing infrastructure (the high school and college courses), does not require close geographic ties, and enrolls a small share of students

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<sup>2</sup>Beginning with the class of 2023, students enrolling full-time at a community college in Michigan within 15 months of high school graduation receive the Community College Guarantee, which awards the full value of in-district tuition and mandatory fees. For out-of-district students, this means that they get discounted, but not free tuition. I restrict the analysis sample to exclude students who were eligible for the Community College Guarantee.

<sup>3</sup>In some cases, EMC students are also offered stand-alone course sections on the campus of their home high school.

still generate positive effects on post-secondary enrollment and attainment?

Based on temporal and geographic variation in the availability of EMCs across Michigan, I estimate the effect of EMC enrollment using a two-stage least squares (2SLS) approach that employs a difference-in-differences specification in each stage and instruments for EMC enrollment with 9th-grade cohort-level EMC eligibility. This way, I compare students who are just young enough to enroll in a new EMC with students who were too old to enroll when the program was first implemented, identifying treatment effects based on the precise timing of program creation rather than the decision to enroll.

I focus on the impacts of these programs on educational attainment. I analyze the number of credits earned at 2-year and 4-year IHEs in each year following 9th grade. By separating 2-year and 4-year credits, I look for possible substitution from 4-year credits to 2-year credits as 2-year credits become relatively cheaper with the availability of Early Middle College. I find that eligibility for Early Middle College both increases and accelerates credit and credential-earning at 2-year IHEs. This increase seems to arise from mild substitution away from enrollment at 4-year IHEs, though there is not a statistically significant impact on Bachelor's degree attainment within five years of high school graduation.

I investigate this substitution more carefully by categorizing students into a set of mutually-exclusive post-secondary pathways: never earning 12 post-secondary credits in any sector; only earning 12 credits at 2-year IHEs; only earning 12 credits at 4-year IHEs; earning 12 credits at 2-year IHEs, then 4-year IHEs; earning 12 credits at 4-year IHEs, then 2-year IHEs; and contemporaneously reaching 12 2-year credits and 12 4-year credits. Early Middle College enrollment drives students away from the 4-year-only path toward the 2-year-only path, with little change among the other post-secondary trajectories. Given that access to Early Middle College increases credit taking and degree attainment at 2-year institutions, with little impact on the likelihood of any degree attainment within 8 years of 9th grade, this suggests that some enrollees may have otherwise earned some 4-year credits, but instead earned an associate degree.

The rest of the paper proceeds as follows. Section 2 outlines the institutional context. Section 3 discusses the data. Section 4 presents the empirical strategy. Section 5 displays the results. Finally, 6 concludes.

## 2 What is Early Middle College?

### 2.1 Institutional Details

Early Middle College (EMC) has spread rapidly across Michigan over the past decade.<sup>4</sup> Students must choose to enroll in an EMC prior to their 11th-grade year. When they enroll, they agree to remain in high school for a fifth year and begin to enroll in college classes at the partner IHE alongside their high school coursework. As they progress through high school, they take a greater share of college classes relative to high school classes until they take nearly a full schedule of college classes during their fifth year. These classes can be in-person on the college campus, online, or in-person within the high school building.

Enrolled students are expected to complete at least one of five program objectives by the end of their fifth year of high school: (1) Earn 60 transferable college credits; (2) Earn an associate degree; (3) Earn a professional certification; (4) Earn a Michigan Early Middle College Association (MEMCA) technical certificate; or (5) Participate in a registered apprenticeship. When students complete the EMC program, they also receive a high school diploma. So, in exchange for their commitment to remain in high school for an additional year, students get the opportunity to earn a significant number of post-secondary credits at no cost.<sup>5</sup>

EMC programs are now widely distributed across Michigan. Figure 1 maps the locations

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<sup>4</sup>Early Middle Colleges can take three forms: in EMC high schools, 100 percent of students are expected to enroll in Early Middle College; in EMC programs, less than 100 percent of students enroll in the EMC; and in EMC consortia, a single program spans multiple school districts. The vast majority of EMCs are either Early Middle College programs or are part of Early Middle College consortia. In my analysis sample, I exclude whole-school EMCs to focus on the small-scale early college model, but the main results are robust to including them.

<sup>5</sup>If students fail EMC college courses, they may be forced to repay the district for any course fees that are not reimbursed by the IHE. This is consistent with the broader dual enrollment policy in Michigan.

of schools that have ever offered an Early Middle College. The green dots represent EMCs, the purple dots represent 2-year IHEs, and the blue dots represent 4-year IHEs. EMC locations seem to be related to the nearby availability of a post-secondary partner and, correspondingly, the areas with denser populations; however, there are EMCs throughout almost all regions of the state, even at schools that are relatively distant from any post-secondary institution.

There has been substantial variation in the timing of Early Middle College formation. Figure 2 shows that the number and share of students enrolled in Early Middle College in Michigan have grown steadily since the 2008-09 school year, with nearly 17,500 students (3.5 percent of high school students) enrolled in 2024-25. EMC enrollment ebbed slightly in 2021-22 and 2022-23 after the COVID-19 pandemic, but resumed its growth in more recent post-pandemic years.<sup>6</sup>

## 2.2 Conceptual Framework

The introduction of a local Early Middle College is contingent upon the existence of a (somewhat) local IHE, so it is important to consider the extent to which the availability of an EMC impacts students' educational choice set. Community colleges already provide a relatively inexpensive option for students looking to earn post-secondary credits relative to 4-year college options. In 2023-24, the average Michigan community college charged about \$4,800 per year in tuition and fees.<sup>7</sup> As of fall 2024, many students in Michigan will have the opportunity to attend tuition-free community college through the Community

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<sup>6</sup>Figure B1 shows the number of active EMC programs and schools by school year. From the initial programs created in 2011-12, nearly 600 schools had active Early Middle Colleges by 2019-20. Despite many of these programs ceasing official operations during the COVID-19 pandemic, the statewide share of students in Early Middle College continued to grow. In my definition of treatment, described in detail in Section 4.1, I do not account for program closure. Therefore, my estimates of the effects of EMC eligibility can be thought of as a lower bound. They include some cohorts as treated, even if their school is no longer operating an EMC program, dulling any potential effects. I choose this definition because it is possible that forming a partnership with an IHE for Early Middle College could have lasting impacts on students' access to dual enrollment.

<sup>7</sup>[https://sfa.senate.michigan.gov/Departments/DataCharts/DCccl\\_TuitionFees.pdf](https://sfa.senate.michigan.gov/Departments/DataCharts/DCccl_TuitionFees.pdf)

College Guarantee program.<sup>8</sup> Still, providing an earlier exposure to college as a realistic option, maintaining the comfort and support of the home high school environment, and the accountability to EMC requirements could lead to improved take-up and persistence relative to standard, post-high school community college enrollment, even if the change in overall cost per credit were relatively small.

As students progress through high school, they can take several potential educational pathways. They choose whether to drop out or complete high school, whether to attempt college credits during high school, whether to go to college or enter the workforce, and whether to attend a 2-year or 4-year IHE. Access to an Early Middle College forces students to think about their future educational plans during their 10th-grade year. At that stage, prior to finalizing any of the decisions described above, EMC-eligible students are asked to decide whether to commit to a fifth year of high school and to enroll in a substantial number of classes at a 2-year college. Choosing this route early on decreases the cost of the first 60 college credits, while guaranteeing that the student will begin their education at a 2-year IHE.<sup>9</sup> The early commitment device could be particularly impactful for high school students with especially high future discount rates. Castillo, Jordan, and Petrie (2019) find that students with higher discount rates are less likely to graduate from high school, documenting that future discounting could play an important role in educational decision making.

Because Early Middle College requires an early commitment relative to other post-secondary choices, it allows students to bypass some future decision points where they may otherwise choose to end their educational progression and join the labor market. However, the early commitment resets expectations away from graduating high school in four years and start immediately at a 4-year institution. This could be a concern if students rush into Early Middle College when they otherwise would have begun their post-secondary studies

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<sup>8</sup><https://www.michigan.gov/mistudentaid/programs/michigan-achievement-scholarship/community-college-guarantee>

<sup>9</sup>There are a limited number of EMCs that partner with 4-year IHEs, including Saginaw Valley State University, University of Michigan-Flint, and Ferris State University.

at a 4-year institution after the usual four years of high school.

Existing literature has documented the possibility that enrollment in 2-year IHEs reduces bachelor's degree attainment based on high attrition rates at 2-year colleges and potential diversion of students from direct routes to 4-year universities. (Pretlow, Cameron, and Jackson 2022; Long and Kurlaender 2009; Monaghan and Attewell 2015) Mountjoy (2022) finds that, for students who would otherwise not obtain any post-secondary education, there is significant value in attending community college. For students who would otherwise have started right away at four-year institutions, beginning at a community college can result in a diversion from future attainment. In 2022-23, the average retention rate in public 2-year institutions was 66.8 percent for students who entered straight from high school, compared to 88.5 percent at public 4-year institutions.<sup>10</sup> Given that EMC makes community college cheaper and more accessible, it is likely that the first-order effect of a new EMC is an increase in the share of students who ever take community college courses. Given the institutional support from the home high school, it is also likely that EMC students have lower attrition rates from community college programs relative to their post-high school counterparts.

The relative shares of students at the margin between community college and no college and the margin between 2-year and 4-year college will dictate the total attainment effects of an EMC program. Both of these marginal groups would be among the most likely EMC enrollees since they would be particularly sensitive to the change in price and support structure that could make 2-year college more desirable.

### 3 Data

For this study, I link two sets of student-level data from the Michigan Education Data Center (MEDC). K-12 student records show enrollment, demographics, course-taking, and graduation as well as EMC participation. post-secondary student records include enrollment, course-taking, and credentials.

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<sup>10</sup><https://nscresearchcenter.org/persistence-retention/>

Together, this allows me to identify students who are eligible for Early Middle College programs during high school and students who enroll in EMCs, and observe post-secondary attainment for both eligible and ineligible students. The post-secondary data covers all students in Michigan since 2011.<sup>11</sup> I rely on students' eighth-grade test scores to control for their pre-EMC academic performance. To allow for observation of students for at least 3 years beyond the standard four-year high school graduation timeline, I include the 2011–2018 9th grade cohorts in my analysis.

### 3.1 Construction of Early Middle College Eligibility

I define student-level EMC eligibility based on the year and school at which a student begins the ninth grade.<sup>12</sup> If at least two students in a ninth-grade cohort at a given school go on to enroll in Early Middle College at that school, then everyone in the cohort is EMC eligible. Otherwise, all students in the cohort are coded as ineligible for EMC enrollment. This creates two types of ineligible cohorts. First, all students who attended 9th grade at schools that never implemented Early Middle College are ineligible. Second, students are ineligible for Early Middle College if they progressed beyond the start of 11th grade by the time their school created an Early Middle College program.

For this second class of students, I define relative treatment time by ninth-grade cohorts. The relative time is zero for the first cohort of ninth graders who are eligible to enroll in that EMC program. For the cohort that is one year older than the first EMC cohort, the relative time is -1. For the cohort that is one year younger than the first cohort, the relative time is +1, and so on.

Figure 3 shows the first-eligible 9th-grade cohort for each Early Middle College program

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<sup>11</sup>Data is most complete for students who attend public IHEs, but National Student Clearinghouse records indicate the university or college enrollment status each semester for students who attend private or out-of-state IHEs. Total credits each semester are also imputed from NSC records. Dynarski, Hemelt, and Hyman (2015) discusses key drawbacks to NSC coverage in more detail.

<sup>12</sup>If students spend multiple years in the ninth grade, I choose their first ninth-grade year. If they attend multiple schools during their first ninth-grade year, then I choose the school that they attended for the most days during that year.

in my analysis sample. I do not include any schools that started EMCs for fall 2013 ninth graders or earlier to maintain at least three pre-treatment periods for every treated school. While over 60 new EMC programs were created for each of the fall 2014 and fall 2015 cohorts, implementation slowed over the rest of the treated sample with between 20 and 30 new EMCs formed for each cohort between fall 2016 and fall 2018.

### **3.2 Which Schools Implement Early Middle College Programs?**

The timing and location of Early Middle College creation is not random. Districts strategically create these programs, potentially due to concerns about low post-secondary attainment or expectations of high demand. Therefore, schools that have EMCs may have different characteristics than schools that have not started EMCs.

Table 1 shows summary statistics for schools that implemented Early Middle College compared to schools that have not implemented Early Middle College. Students in schools are relatively higher scoring on both 8th grade and the SAT. Early Middle College programs are also more likely to be in suburbs and rural areas and less likely to be in urban areas than schools without Early Middle College. EMC implementing schools also have a higher share of white students and lower share of Black students than schools that never implemented EMC. Finally, EMC schools have lower shares of students classified as English Language Learners or economically disadvantaged. This suggests that relatively higher achieving, higher income students have more access to Early Middle College. This is generally consistent with the fact that EMCs are at least loosely geographically tied to institutions of higher education. Given these differences, I show that the main results are robust to using only not-yet treated schools as controls instead of including never-treated schools.

### **3.3 Which Students Enroll in Early Middle College Programs?**

In addition to the selection of high schools into EMC provision, there is an endogenous selection of students into the EMC at their high school. Table 2 shows that about 3.5

percent of students enroll in EMC in the first eligible cohort at each participating school. EMC enrollees are more likely to be white, more likely to be female, less likely to be English Learners, and less likely to be in special education than students who do not enroll. EMC students also tend to have higher levels of past academic achievement based on their 8th-grade test scores. This implies that they likely have higher academic ability and drive before program exposure. Because of these differences, direct comparisons between enrollees and non-participants would likely be biased. Instead, my identification strategy is based on the timing of EMC formation and cohort-level eligibility rather than comparing outcomes for students who enroll to those who do not enroll.

### **3.3.1 Comparison Between Completers and Non-Completers**

I denote students as Early Middle College enrollees if they are ever reported as Early Middle College students. This means that the results reported in Section 5 implicitly account for attrition from Early Middle College programs. In my analysis sample, 46.8 percent of EMC enrollees graduate with their high school diploma and an EMC-approved credential after five years of high school. This relatively low completion rate could lead to underestimates of the impacts from program completion in my framework.

Table 3 shows group-level averages conditional on HS graduation status, comparing on-time 4-year HS graduates who were eligible for Early Middle College but did not enroll to Early Middle College completers and Early Middle College non-completers. Early Middle College non-completers enrolled in EMC at some point during high school but either graduated with an EMC credential after more than five years or did not graduate with an EMC credential. By comparing Columns 2 and 3, it is clear that non-completers are far less likely to go on to earn post-secondary credentials and disproportionately likely to be lower-achieving and economically disadvantaged compared to program completers. This could mean that there is scope for increased institutional support from the home high school for EMC students, particularly for more disadvantaged students.

Figures 4 and B4 plot the credit earning trajectories for EMC completers and enrollees who attrit from the program from 9th grade through four years after 12th grade. Both groups of EMC enrollees accumulate substantially more college credits than their peers during high school. Naturally, these dual enrollment credits are almost exclusively earned at 2-year colleges. For enrolled non-completers, credit attainment stalls after the traditional timing of high school graduation. For completers, average attainment intensifies each year from the conventional junior year of high school through the fifth and final high school EMC year resulting in about 50 total credits at the point of program completion, 4 years after 9th grade. This is consistent with the stated goals of Early Middle College for students to earn an associate degree, credential, or 60 transferable credits. However, on average, neither EMC completers nor non-completers are particularly likely to continue earning two-year credits or to transfer to a four-year IHE.

Students who begin their post-secondary studies at 4-year institutions overtake EMC completers' total credit attainment about two years after high school graduation. Students who do not enroll in EMC, but instead initially enroll at 2-year colleges after graduating from high school take four full years after high school graduation to earn the amount of credits that EMC students earn during high school.

### 3.3.2 EMC Students vs. Students on Other Post-secondary Pathways

Table A1 compares EMC enrollees to students from EMC eligible cohorts who followed various post-secondary pathways. On average, EMC students have higher standardized test scores than students who never enroll in multiple post-secondary courses in a single fall or spring semester after their high school graduation and students whose initial enrollment after graduation is at a 2-year institution. EMC students score lower than students who begin at 4-year private, 4-year public, or 4-year flagship IHEs.<sup>13</sup> EMC enrollees have similar rates of economically disadvantaged status to 2-year enrollees, but higher rates than students who

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<sup>13</sup>In Michigan, I define the University of Michigan and Michigan State University as flagship IHEs.

go on to initially enroll at 4-year institutions.

## 4 Empirical Strategy

I estimate the effects of Early Middle College on student-level educational attainment using a 2-stage least squares design with a difference-in-difference (DID) specification at each stage. First, I define the treatment variable based on cohort-level program eligibility. Then, in the first stage, I predict program take-up based on program eligibility. Finally, I estimate the impacts of predicted take-up on several educational outcomes and show event studies for the reduced form effects of EMC eligibility on academic outcomes.

### 4.1 Construction of Early Middle College Eligibility

I define student-level EMC eligibility based on the year and school at which they begins the ninth grade.<sup>14</sup> If at least two students in a ninth-grade cohort at a given school go on to enroll in Early Middle College at that school, then everyone in the cohort is EMC eligible. Otherwise, all students in the cohort are coded as ineligible for EMC enrollment. This creates two types of ineligible cohorts. First, all students who attended 9th grade at schools that never implemented Early Middle College are ineligible. Second, students are ineligible for Early Middle College if they progressed beyond the start of 11th grade by the time their school created an Early Middle College program. In Appendix Tables A6 through A8, I show that my results are robust to using administratively-reported school-level program start dates instead.<sup>15</sup>

For this second class of students, I define relative treatment time by ninth-grade cohorts. The relative time is zero for the first cohort of ninth graders who are eligible to enroll in that

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<sup>14</sup>If students spend multiple years in the ninth grade, I choose their first ninth-grade year. If they attend multiple schools during their first ninth-grade year, then I choose the school that they attended for the most days during that year.

<sup>15</sup>I prefer the student-based definition because it does not rely on assumptions about the oldest grade eligible to enroll in the Early Middle College when it is implemented. For example, some schools might require enrollment in 10th grade, while others require enrollment in 11th grade. For the school-reported sample, students are coded as eligible if their school began offering an EMC prior to them reaching 11th grade.

EMC program. For the cohort that is one year older than the first EMC cohort, the relative time is -1. For the cohort that is one year younger than the first cohort, the relative time is +1, and so on.

Figure 3 shows the first-eligible 9th-grade fall year for each Early Middle College program in my analysis sample. While it was most common for schools to form new EMCs for the fall 2014 and fall 2015 cohorts of 9th-grade students, new program formation continued throughout the next four years. I do not include any EMCs that had eligibility prior to the fall 2014 incoming 9th graders to allow for three pre-treatment cohorts.

#### 4.1.1 Covariate Balance

By estimating the effects of Early Middle College eligibility instead of Early Middle College enrollment, my approach avoids concerns about the endogeneity of program enrollment. Still, the timing of program implementation could also be endogenously determined and driven by changes in either observable or unobservable student characteristics. Alternatively, it is possible that students relocate into a school for the new Early Middle College program. However, this is less likely to impact my results because students would have to relocate for 9th grade, two years in advance of when they would join the future EMC to be included in the analysis sample.

Table A2 shows that the demographics and test scores of the last ineligible cohort and the first eligible cohort at EMC-implementing schools are very similar. This suggests that there is no anticipatory sorting of students and that the precise timing of implementation is not related to changes in observable student characteristics between cohorts.

## 4.2 First Stage

In the first stage, I model the relationship between EMC eligibility and EMC enrollment as follows:

$$EMCenroll_{ist} = \alpha + \beta EMCeligible_{st} + \lambda X_i + \delta_t + \gamma_s + u_{ist}$$

$EMCenroll_i$  is an indicator for whether student  $i$  ever enrolls in an Early Middle College during high school.  $EMCeligible_{ts}$  is an indicator for whether a student is eligible for Early Middle College given their 9th-grade cohort year  $t$  and their 9th-grade school  $s$ .  $X_i$  is a vector of time-constant student characteristics, including 8th-grade math scores from state standardized tests and indicators for students' race, sex, economically disadvantaged status, English learner status, disability status, and special education status.  $\delta_t$  and  $\gamma_s$  are cohort-year and cohort-school fixed effects.

Figure 5 shows the results from the first stage as an event study based on the following specification:

$$EMCenroll_{ist} = \alpha + \sum_{\tau=-3, \tau \neq -1}^2 \left( \beta_t * EMCeligible_s * 1(Cohort = \tau) \right) + \lambda X_i + \delta_t + \gamma_s + u_{ist}$$

$\beta_t$  is the estimated relationship between EMC eligibility and EMC enrollment for cohort  $t$ , the cohort that started 9th grade  $t$  years after the first cohort eligible for the EMC.

Note that based on my definition of eligibility, it is not possible for ineligible students to enroll in the Early Middle College at their 9th-grade school; however, it is possible for them to enroll in an EMC if they move to a high school with an EMC at some point between 9th and 11th grade. The lack of pre-trend in Figure 5 shows that few students pursue an EMC by relocating after the start of high school. There is a clear discontinuous jump in the likelihood of EMC enrollment for the first cohort that is eligible, with the estimated take-up rate climbing from just under 3 percent in the initial cohort to just over 3.5 percent by the third eligible cohort. This is consistent with Early Middle College utilization increasing as programs become more established at a school.

### 4.3 Second Stage

With a clear first-stage relationship between EMC eligibility and EMC enrollment, I estimate the impact of predicted EMC enrollment on outcomes rather than using EMC enrollment

itself. The second stage takes the following form:

$$Y_{ist} = \alpha + \beta \widehat{EMCenroll}_{ist} + \lambda X_i + \delta_t + \gamma_s + u_{ist}$$

$\widehat{EMCenroll}_{ist}$  is the predicted EMC enrollment status from the first stage.  $Y_{ist}$  is the outcome of interest. The Wald estimator,  $\frac{\widehat{\beta_{reduced-form}}}{\beta_{first-stage}}$  gives the estimated effect of EMC enrollment on each outcome. Importantly, if the provision of an Early Middle College impacts educational decisions among non-enrollees by providing more information about two-year post-secondary options or based on peer interactions, the 2SLS estimates could overstate the effects for program enrollees. Given the low take-up rates for Early Middle College, I do not think that there are large spillovers to non-participants; however, to the extent that there are, it is more natural to interpret the reduced-form estimates.

#### 4.4 Reduced Form Event Study

To more clearly observe pre-trends, I also plot reduced-form event studies. These are estimated comparably to the first-stage event study:

$$Y_{ist} = \alpha + \sum_{\tau=-3, \tau \neq -1}^2 \left( \beta_t * EMCeligible_s * 1(Cohort = \tau) \right) + \lambda X_i + \delta_t + \gamma_s + u_{ist}$$

Since the treatment variable is EMC eligibility rather than EMC enrollment, the treatment effect measured here does not account for the take-up rate among EMC-eligible students. Given that Figure 5 shows take-up rates of about 3 percent, the reduced form estimates are scaled up considerably when interpreting them relative to actual EMC enrollment in two-stage least squares. I also show all event studies estimated using the extended two-way fixed effects (ETWFE) approach from Wooldridge (2025) which is robust to staggered treatment timing and heterogeneous treatment effects. Generally, the estimates from TWFE and ETWFE are very similar for each outcome.

## 5 Results

Early Middle College operates as a hybrid between high school enrollment and post-secondary enrollment, with a greater prioritization on post-secondary course-taking than conventional dual enrollment. Therefore, I consider the impacts of EMC eligibility on high school students and on attainment up to seven years following 9th-grade entry. For students who enroll in Early Middle College, this represents their full high school experience and three post-secondary years. For non-enrollees, this includes their four years of high school and four post-secondary years.

### 5.1 High School Outcomes

Enrolling in Early Middle College during high school changes students' experience in several key ways. Enrolled students graduate in five years instead of four years. Additionally, students in EMC take community college courses throughout their high school years, potentially exposing them to a different set of curricula, academic supports, and peers that could impact their academic achievement and future enrollment trajectory.

#### 5.1.1 Dual Enrollment

Early Middle College is a dual enrollment program, with enrolled students pursuing 60 transferrable college credits or an associate degree. Because it is very uncommon for a high school student to enroll in that many college credits in the absence of an EMC, EMC students almost certainly earn more college credits in the course of their high school career than they would have in the absence of the program.

Figure 6 plots the effect of Early Middle College eligibility on total post-secondary credit accumulation between 9th grade and 12th grade. For non-EMC students, this measures their total dual enrollment course-taking. For EMC students, this covers two of their three main years of EMC enrollment. There is a clear discontinuous increase in dual enrollment credits

earned during the standard high school years after schools implement Early Middle College. This means that Early Middle College is not substituting for dual enrollment that would have happened otherwise. Figure B2 separates this effect into credits earned at 2-year institutions and credits earned at 4-year institutions. It is clear that the effect comes from an increase in 2-year credit attainment, with little to no impact on dual enrollment in courses at 4-year colleges. This suggests that Early Middle College enrollment does not induce substitution away from dual enrollment at 4-year universities.

The 2-stage least squares estimates in Table 4 show that, compared to their peers, EMC enrollees earn an additional 15.9 credits at 2-year IHEs by the end of their fourth year of high school. This difference represents an 870 percent increase in credits relative to the pre-treatment mean at treated schools and combines with a modest increase in dual enrollment at four-year IHEs (1.4 credits) to give students an additional 17.3 college credits at the end of the conventional high school timeline, an effect of 779 percent. Figure 7 shows that the disparity in 2-year credits grows during the final EMC year (4 years from 9th grade), while non-EMC students begin to earn more 4-year credits than their EMC peers who remain in high school at that point.

### 5.1.2 Graduation and Achievement

Early Middle College could impact students' high school outcomes based on increased exposure to dual enrollment or alter the likelihood of 4-year and 5-year graduation based on its program requirements. However, Table 4 shows no evidence of impacts on 4-year graduation rates, 5-year graduation rates, or SAT scores after EMC implementation. Given the low take-up rates for Early Middle College and the high attrition rate from EMC programs, it is not surprising that it has little impact on aggregate graduation rates.

## 5.2 Post-secondary Outcomes

Next, I analyze the impacts of Early Middle College on students' educational attainment beyond their senior year of high school. I consider the decision to ever enroll in post-secondary coursework, the choice of post-secondary pathway, credit earning behavior, and degree attainment.

### 5.2.1 Enrollment

Eligibility for Early Middle College affects the timing and choice set for students' decisions to enroll in post-secondary courses. EMC eligible students have new access to free courses at a 2-year IHE and decide whether to enroll prior to their junior year of high school. However, Table 5 shows that there are no statistically significant impacts on whether enrollees ever enroll at 2-year or 4-year institution within 7 years<sup>16</sup> after entering 9th grade, though the coefficient on 2-year enrollment is positive while the coefficient on 4-year enrollment is negative. Figure B3 plots the reduced form relationship between EMC eligibility and any post-secondary enrollment and shows no evidence of differential pre-trends between the treated and control groups and some evidence of a positive impact for the third treated cohort. While a lack of prominent effects on the extensive margin may seem surprising, over 70 percent of students at treated schools in the cohort prior to treatment go on to attend some college, so for a majority of students, there is only room to boost attainment on the intensive margin.

### 5.2.2 Pathways

Table 6 compares the effects of Early Middle College enrollment on students' choice to follow one of six mutually exclusive post-secondary pathways: not reaching 12 credits at 2-year nor

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<sup>16</sup>I focus on 7 years for most estimates because it balances having more of the sample of treated schools available with also capturing several years of post-secondary enrollment and course-taking decisions after high school. Normal high school graduation is 3 years after 9th grade, so this captures high school, the extra EMC year, and an additional 3 years after the completion timing for EMC.

4-year IHEs; only reaching 12 credits at 2-year IHEs; only reaching 12 credits at 4-year IHEs; reaching 12 credits at 2-year IHEs, then reaching 12 credits at 4-year IHEs; reaching 12 credits at 4-year IHEs, then reaching 12 credits at 2-year IHEs; and reaching 12 credits contemporaneously at 2-year and 4-year IHEs.<sup>17</sup> These outcomes are measured 7 years after 9th grade which is 4 years after standard high school graduation.

Early Middle College seems to encourage students to move from only reaching 12 credits at 4-year IHEs to only reaching 12 credits at 2-year IHEs, rather than making it less likely that students will ever reach 12 credits in either sector. EMC enrollment leads to a 22.8 percentage point increase in the 2-year only path and a corresponding 23.5 percentage point decrease in the 4-year only path. There is a positive, not statistically significant effect on transferring from 2-year IHE to 4-year IHE, suggesting that students pushed toward 2-year enrollment by EMC eligibility are at least as likely to transfer on to a 4-year institution as other non-EMC enrollees at 2-year IHEs.

Early Middle College accelerates the timeline for earning at least 12 credits at 2-year institutions. Among students who successfully reach the 12-credit milestone at a 2-year institution, EMC enrollees do so 1.4 years earlier than non-enrollees. Because EMC increases the overall likelihood of reaching this threshold, this difference reflects both accelerated course-taking and changes in the composition of students reaching the milestone. Although EMC enrollees who attain both the 2-year and 4-year credit thresholds take more time to matriculate to 4-year institutions after earning 12 2-year credits, their substantial head start in earning 2-year credits during high school generally allows them to reach the 12-credit milestone at a 4-year institution earlier than traditional transfer students.

### 5.2.3 Credit Earning

Given the substitution between 2-year and 4-year enrollment, I next turn to the impact on overall credit earning and degree attainment for broader understanding of the way in which

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<sup>17</sup>This last category is omitted from the results tables for brevity. It is uncommon, difficult to interpret theoretically, and the estimated coefficient is small and not statistically significant across specifications.

EMC enrollment changes post-secondary decision making. Figure 7 plots the two-stage least squares effect of EMC enrollment on credit taking within each year range from 9th grade (0 years from 9th grade) through 8 years after 9th grade (5 years after conventional high school graduation).<sup>18</sup> It is clear that during the Early Middle College years, 2 through 4 on the graph, Early Middle College is causing students to take substantially more 2-year college credits. During the high school years, there is no evidence of substitution away from 4-year credit taking. However, during EMC students' 5th year of high school they continue to earn more 2-year credits, but earn significantly fewer 4-year credits as they are still tied to their EMC commitments. The ensuing trajectory over the four remaining analysis years suggests that non-EMC students are not achieving the same level of 2-year attainment as their EMC peers, even after leaving high school.

The gap in credit attainment at 4-year institutions seems to plateau in years 7 and 8 after 9th grade at about 35 fewer credits for EMC students. Figure B5 shows the reduced form event study for the effect on credits through year 7, by type. There is no evidence of differential pre-trends for either 2-year or 4-year credit taking.

#### 5.2.4 Degree Attainment

Figure 8 plots the reduced form effect of EMC eligibility on degree attainment for associate degree (within 4 years of conventional high school graduation) and bachelor's degree (within 5 years of conventional high school graduation). Early Middle College eligibility causes students to earn more associate degrees. There appears to be a slight negative, though not statistically significant impact on bachelor's degree attainment. The two-stage least squares coefficients in Table 5 show that EMC students are 4.8 times more likely to earn an associate degree. These estimates suggest that enrollees are about 75 percent less likely to earn a bachelor's degree on this timeline, but that coefficient is imprecisely estimated and

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<sup>18</sup>To facilitate comparisons between time horizons, the sample underlying this figure is restricted to include only students for whom I can observe the full 8 years after 9th grade. This allows for a consistent sample for each estimate.

not statistically significant. Combined, this leads to a statistically significant increase in the rate of earning any degree within 4 years of conventional high school graduation, but no effect on degree attainment within 5 years of conventional high school graduation timing.<sup>19</sup>

The bachelor’s result should be interpreted with caution because students who transfer on the way to their 4-year degree generally take longer to graduate. For example, in Texas, 2-year to 4-year transfers take an average of 7.3 years to graduate from first enrollment versus 5.2 years for non-transfer students. (Texas Higher Education Coordinating Board 2023) This suggests that 5 years after standard high school graduation is too soon to observe most bachelor’s degree attainment among students who are transferring from a 2-year institution. With a restricted analysis sample, Figure B9 shows suggestive evidence that, over longer time horizons (up to 7 years beyond the standard high school graduation timeline), EMC enrollees seem to catch back up to their peers in bachelor’s degree attainment. This is consistent with a story where Early Middle College delays but does not replace bachelor’s degree attainment. Further analysis over a longer time horizon is needed to provide a definitive conclusion about the impacts of Early Middle College on bachelor’s degree attainment.

### 5.3 Heterogeneity

Figures B6 through B8 explore heterogeneity in the effect of EMC enrollment on post-secondary enrollment, credit earning, and degree attainment by race, sex, income level, and high school urbanicity. While these models do not formally test for equality across groups, the point estimates suggest varying patterns by race, sex, and income level. The estimated impacts on 2-year enrollment and course taking are smaller for Black students, who instead show more positive point estimates for 4-year enrollment, credit taking, and degree attainment. Non-economically disadvantaged students show large increases in 2-year enrollment, credits, and associate degree attainment, whereas the estimates for economically disadvantaged students are small and not statistically significant.

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<sup>19</sup>The results in Table A9 show that this finding is not a result of differences in sample composition, it is robust to restricting to a sample with both the 7-year and 8-year horizon available.

Increases in associate degree attainment appear most prominent among students at suburban schools. As these differences may reflect differing program take-up rates or distinct complier populations across subgroups, these patterns are descriptive rather than causal. However, they suggest it is important to consider the local demographic context when forecasting the impact of a newly-formed Early Middle College.

## 5.4 Robustness

In addition to showing robustness to using extended two-way fixed effects for all reduced form results, Tables A3 through A5 show that the main results are robust to restricting to a balanced sample. I do this in two ways. In the middle column of each table, I show the results restricting to only the first post-treatment period. In the third column, I restrict treated schools to only those that have all three post-treatment cohorts.

Tables A6 through A8 assign treatment based on official program start dates instead of observed student enrollment, potentially allowing for broader coverage of EMCs with particularly low enrollment. With this eligibility definition, results look very similar to the main estimates with the exception that I find a significant negative effect of EMC enrollment on the likelihood of never reaching 12 post-secondary credits and no statistically significant evidence of a change in the likelihood of reaching 12 4-year credits or the number of 4-year credits earned within 7 years after 9th grade.

Tables A9 and A10 show that comparisons between time horizons are robust to using a consistent sample for each estimate.

Finally, Tables A11 through A13 show that the results are robust to only using not-yet treated cohorts as control observations instead of including cohorts from never-treated schools.

## 6 Discussion

While the early college model has proven effective at increasing attainment where it is highly-demanded and closely tied to a post-secondary institution, less is known about the impacts of smaller-scale programs that provide intensive dual enrollment with low take-up and without the same level of connection to a college. Early Middle College programs in Michigan are an example of this sort of small-scale early college model.

Using cohort-level variation in program eligibility, I show that Early Middle College changes the post-secondary decision-making of eligible students. EMC students earn associate degrees faster and more frequently and have accumulated more credits at 2-year institutions as many as 8 years after 9th grade. By exploring substitution between various post-secondary pathways, I show that the program shifts the aggregate distribution of student trajectories, substantially reducing the share of students who accumulate credits only at 4-year institutions and increasing the share who do so only at 2-year institutions. Therefore, Early Middle College accelerates overall credential attainment and increases associate degree completion, with a substitution of 2-year credits in place of 4-year credits.

It can take many years after high school graduation for students to reach their maximum lifetime educational attainment, so future work should explore this question on a longer time horizon. This is particularly relevant for outcomes related to bachelor's degree attainment because students who pursue a 2-year to 4-year transfer to earn a bachelor's degree often take longer to graduate. There may be additional 4-year enrollment for EMC students that has simply been delayed by their choice of a 2-year pathway, rather than crowded out entirely.

Together, the evidence provided here suggests that small-scale early college programs can have important impacts on students' educational trajectories even with minimal infrastructure investment and low enrollment shares.

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Table 1: Comparison of EMC Schools vs. Non-EMC Schools

|                            | Non-EMC School | EMC School |
|----------------------------|----------------|------------|
| Male                       | 0.50           | 0.51       |
| White                      | 0.58           | 0.76       |
| Black                      | 0.27           | 0.10       |
| Hispanic                   | 0.08           | 0.07       |
| Asian                      | 0.04           | 0.03       |
| Other                      | 0.04           | 0.04       |
| English Language Learner   | 0.06           | 0.03       |
| Economically Disadvantaged | 0.49           | 0.39       |
| Special Education          | 0.14           | 0.12       |
| 8th Grade Math             | -0.06          | 0.14       |
| 8th Grade ELA              | -0.07          | 0.11       |
| Urban                      | 0.34           | 0.13       |
| Suburb                     | 0.38           | 0.46       |
| Town                       | 0.09           | 0.12       |
| Rural                      | 0.19           | 0.29       |
| SAT/ACT Composite          | 0.01           | 0.11       |
| N                          | 29,384         | 34,005     |

For EMC schools, these statistics are from the first 9th grade cohort eligible for their Early Middle College. For non-EMC schools, these statistics are from the 2016-17 9th grade cohort, which aligns with the average implementation timing in the analysis sample.

Table 2: Comparison of EMC Students vs. Non-EMC Students

|                            | Non-EMC Students | EMC Students |
|----------------------------|------------------|--------------|
| Male                       | 0.51             | 0.48         |
| White                      | 0.75             | 0.78         |
| Black                      | 0.10             | 0.11         |
| Hispanic                   | 0.07             | 0.06         |
| Asian                      | 0.03             | 0.01         |
| Other                      | 0.04             | 0.04         |
| English Language Learner   | 0.03             | 0.02         |
| Economically Disadvantaged | 0.40             | 0.41         |
| Special Education          | 0.12             | 0.07         |
| 8th Grade Math             | 0.12             | 0.15         |
| 8th Grade ELA              | 0.12             | 0.24         |
| ACT/SAT Composite          | 0.10             | 0.09         |
| Urban                      | 0.14             | 0.07         |
| Suburb                     | 0.47             | 0.39         |
| Town                       | 0.13             | 0.17         |
| Rural                      | 0.26             | 0.37         |
| N                          | 33,367           | 1,243        |

These students were entering 9th graders in the first cohort that was eligible for EMC enrollment at their school.

Table 3: Characteristics and Postsecondary Attainment, by EMC status

|                              | HS Graduates | EMC Cert | EMC No Cert | All EMC |
|------------------------------|--------------|----------|-------------|---------|
| SAT Composite Score (sd)     | 0.16         | 0.33     | -0.01       | 0.15    |
| 8th Grade Math Score (sd)    | 0.21         | 0.40     | 0.11        | 0.24    |
| 8th Grade Reading Score (sd) | 0.21         | 0.49     | 0.18        | 0.32    |
| Economically Disadvantaged   | 0.34         | 0.33     | 0.45        | 0.39    |
| Special Education            | 0.08         | 0.03     | 0.06        | 0.05    |
| English Language Learner     | 0.03         | 0.02     | 0.02        | 0.02    |
| Enrollment after HS          | 0.67         | 0.72     | 0.52        | 0.61    |
| 4-year IHE                   | 0.65         | 0.67     | 0.57        | 0.61    |
| 4-year Public IHE            | 0.51         | 0.57     | 0.42        | 0.49    |
| 4-year Private IHE           | 0.13         | 0.10     | 0.15        | 0.13    |
| 2-year IHE                   | 0.35         | 0.33     | 0.43        | 0.39    |
| State Flagship IHE (UM/MSU)  | 0.16         | 0.10     | 0.09        | 0.09    |
| Bach Degree within 8 Years   | 0.23         | 0.21     | 0.11        | 0.16    |
| Assoc Degree within 5 Years  | 0.00         | 0.49     | 0.04        | 0.25    |
| Assoc Degree within 6 Years  | 0.02         | 0.57     | 0.09        | 0.31    |
| Assoc Degree within 7 Years  | 0.05         | 0.62     | 0.13        | 0.36    |
| Assoc Degree within 8 Years  | 0.07         | 0.65     | 0.17        | 0.39    |
| N                            | 78,362       | 1,985    | 2,324       | 4,309   |

Students who entered high school in EMC eligible cohorts from fall 2014–fall 2018. Column 1 includes students who graduated from high school in four years and never enrolled in EMC. Column 2 includes students who enrolled in EMC and graduated within five years of entry with a HS diploma and postsecondary credential that meets EMC criteria. Column 3 includes students who enrolled in EMC at some point during high school, but did not graduate within five years with an EMC credential. Column 4 includes all students from Columns 2 and 3. Enrollment after HS and degree-attainment rates are shares of the full column sample. IHE-type shares are conditional on post-high-school enrollment; the four-year public and private rows subdivide the four-year category.

Table 4: Effects of EMC Enrollment on High School Outcomes

| Outcome                                 | Coefficient | SE      | N       |
|-----------------------------------------|-------------|---------|---------|
| 2-year credits by end of 4th year of HS | 15.938***   | (2.744) | 406,037 |
| 4-year credits by end of 4th year of HS | 1.371*      | (0.812) | 406,037 |
| Total credits by end of 4th year of HS  | 17.309***   | (2.805) | 406,037 |
| 4-year high school graduation           | -0.056      | (0.096) | 404,984 |
| 5-year high school graduation           | -0.024      | (0.084) | 405,226 |
| 6-year high school graduation           | -0.028      | (0.081) | 405,235 |
| SAT Composite                           | -0.320      | (0.283) | 232,489 |

Standard errors are clustered at the school level. Estimates are based on a 2SLS specification controlling for 8th grade math and reading scores, economically disadvantaged status, special education status, English language learner status, race, and sex. Both the first and second stage regressions include school and year fixed effects. Outcomes of the form X within T years are defined as follows: 0 years means during 9th grade, 3 years means during the standard 4-year high school enrollment, etc. \* p<0.10; \*\* p<0.05; \*\*\* p<0.01.

Table 5: Effects of EMC Enrollment on Postsecondary Attainment

| Outcome                                     | Coefficient | SE      | N       |
|---------------------------------------------|-------------|---------|---------|
| Any postsecondary enrollment within 7 years | 0.090       | (0.129) | 369,834 |
| Any 2-year enrollment within 7 years        | 0.016       | (0.174) | 369,834 |
| Any 4-year enrollment within 7 years        | -0.190      | (0.116) | 369,834 |
| Any degree within 7 years                   | 0.215***    | (0.051) | 369,834 |
| Any degree within 8 years                   | -0.032      | (0.104) | 322,525 |
| Associate degree within 5 years             | 0.204***    | (0.028) | 406,037 |
| Associate degree within 6 years             | 0.215***    | (0.035) | 406,037 |
| Associate degree within 7 years             | 0.219***    | (0.048) | 369,834 |
| Associate degree within 8 years             | 0.158***    | (0.060) | 322,525 |
| Bachelor's degree within 8 years            | -0.145      | (0.098) | 322,525 |

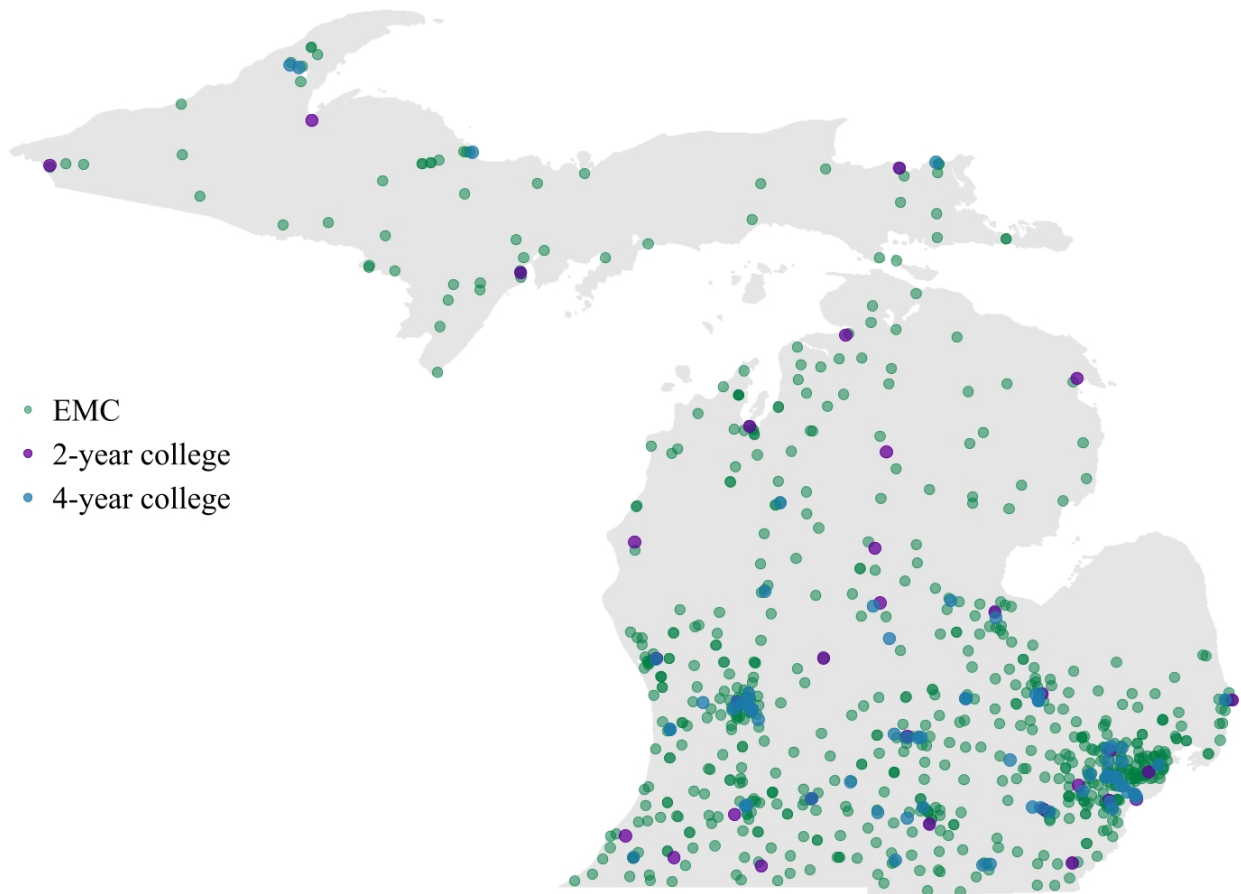
Standard errors are clustered at the school level. Estimates are based on a 2SLS specification controlling for 8th grade math and reading scores, economically disadvantaged status, special education status, English language learner status, race, and sex. Both the first and second stage regressions include school and year fixed effects. Outcomes of the form X within T years are defined as follows: 0 years means during 9th grade, 3 years means during the standard 4-year high school enrollment, etc. \* p<0.10; \*\* p<0.05; \*\*\* p<0.01.

Table 6: Effects of EMC Enrollment on Postsecondary Pathway and Transfer Timing

| Outcome                                       | Coefficient | SE      | N       |
|-----------------------------------------------|-------------|---------|---------|
| Never reached 12 postsec credits              | -0.050      | (0.079) | 270,756 |
| Only reached 12 2-year credits                | 0.228***    | (0.078) | 270,756 |
| Only reached 12 4-year credits                | -0.235**    | (0.111) | 270,756 |
| Reached 12 2-year, then 12 4-year credits     | 0.110       | (0.076) | 270,756 |
| Reached 12 4-year, then 12 2-year credits     | -0.045      | (0.032) | 270,756 |
| Years to 12 2-year credits                    | -1.353***   | (0.247) | 98,440  |
| Years to 12 4-year credits                    | 0.026       | (0.310) | 164,995 |
| Years to 12 total credits                     | -1.117***   | (0.221) | 224,084 |
| Years between 12 2-year and 12 4-year credits | 0.942***    | (0.246) | 29,062  |

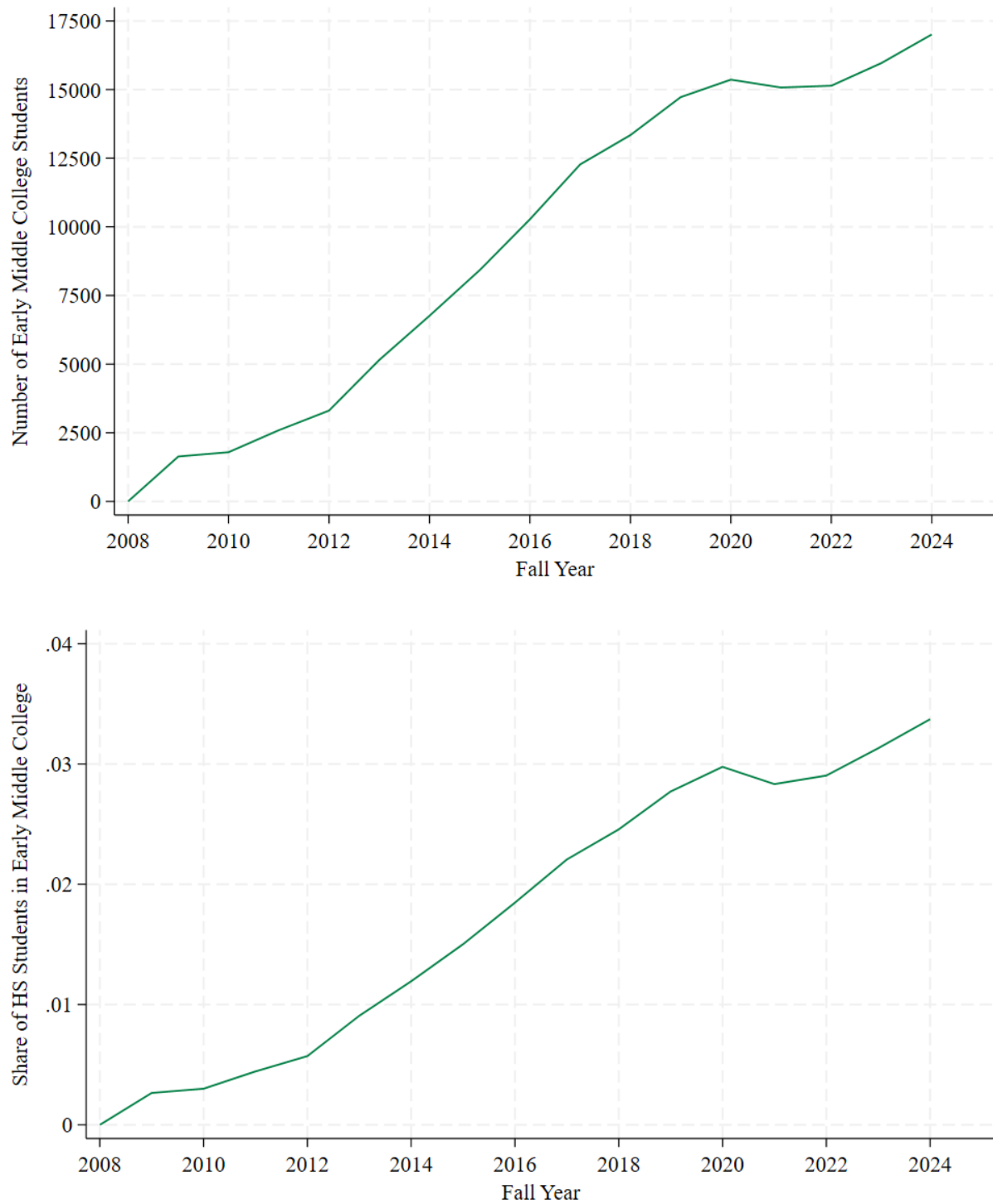
Standard errors are clustered at the school level. Estimates are based on a 2SLS specification controlling for 8th grade math and reading scores, economically disadvantaged status, special education status, English language learner status, race, and sex. Both the first and second stage regressions include school and year fixed effects. A sixth pathway—contemporaneously reaching 12 2-year and 12 4-year credits—was estimated but is omitted for brevity because the effect was small and statistically insignificant. \*  $p < 0.10$ ; \*\*  $p < 0.05$ ; \*\*\*  $p < 0.01$ .

Figure 1: Locations of EMCs and Post-secondary Institutions



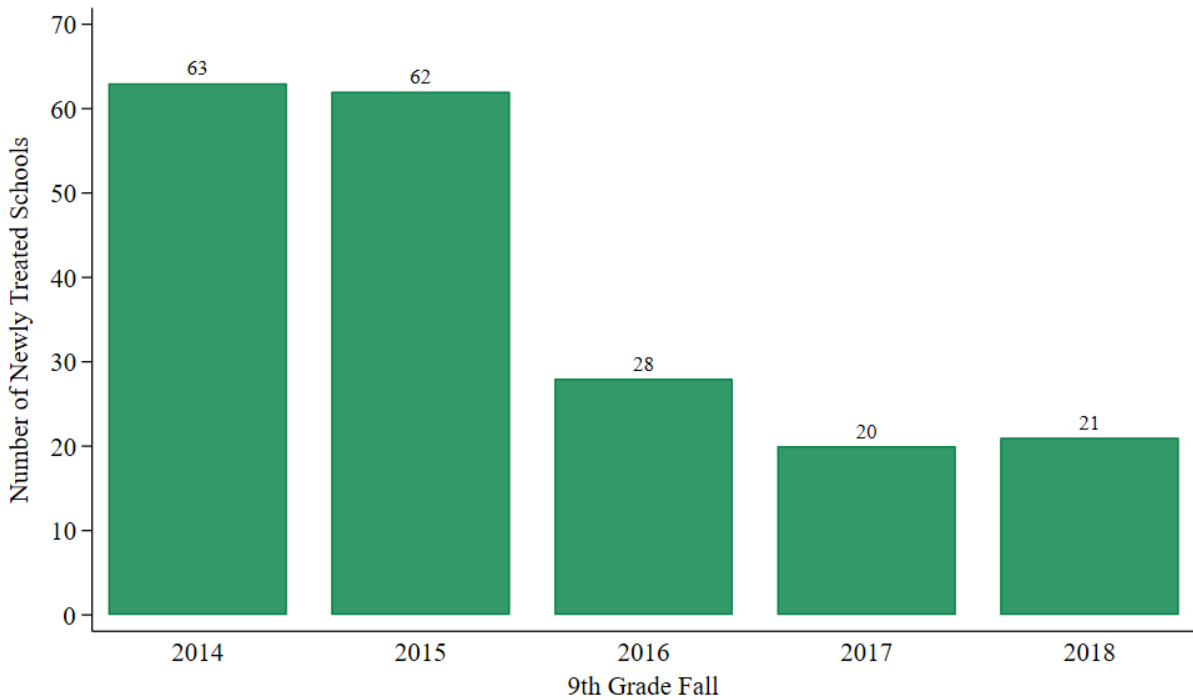
Early Middle College dots represent any school that ever reported having an EMC program or being an EMC school between 2005-06 and 2024-25.

Figure 2: Students Actively Enrolled in Early Middle College



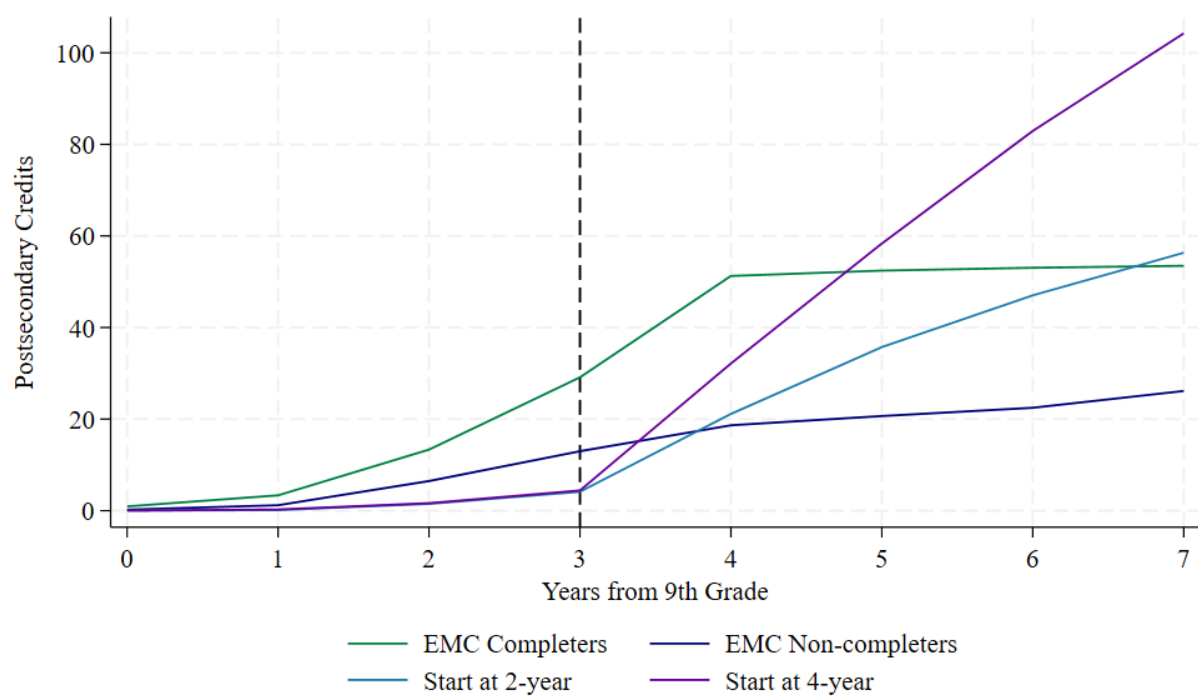
The bottom panel shows the share of students in grade 9–12 who are reported to be enrolled in Early Middle College each school year.

Figure 3: Analysis Sample: Number of New EMC Programs, by 9th Grade Fall Cohort



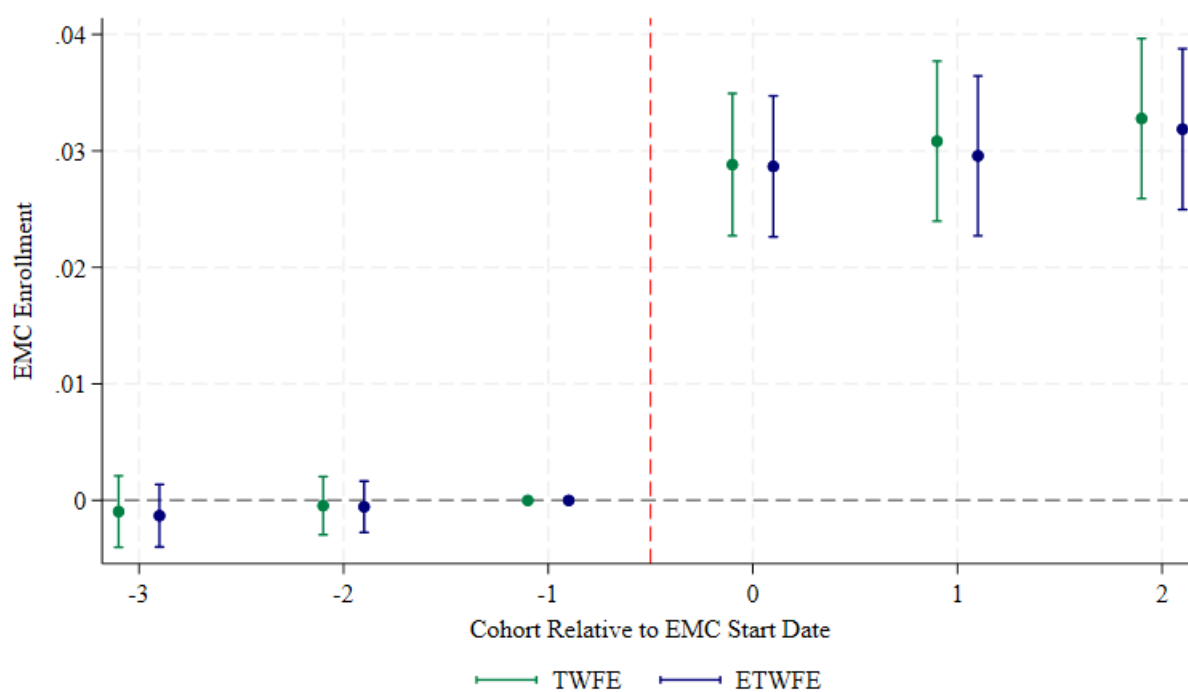
Schools are coded as having a new EMC program for the first 9th-grade cohort that has at least two students go on to enroll in EMC at that school. For the purposes of analysis, all following cohorts of 9th-graders are coded as treated cohorts, even if the EMC ceased operations during or after the COVID-19 pandemic.

Figure 4: Average Total Postsecondary Credit Attainment, by Postsecondary Pathway



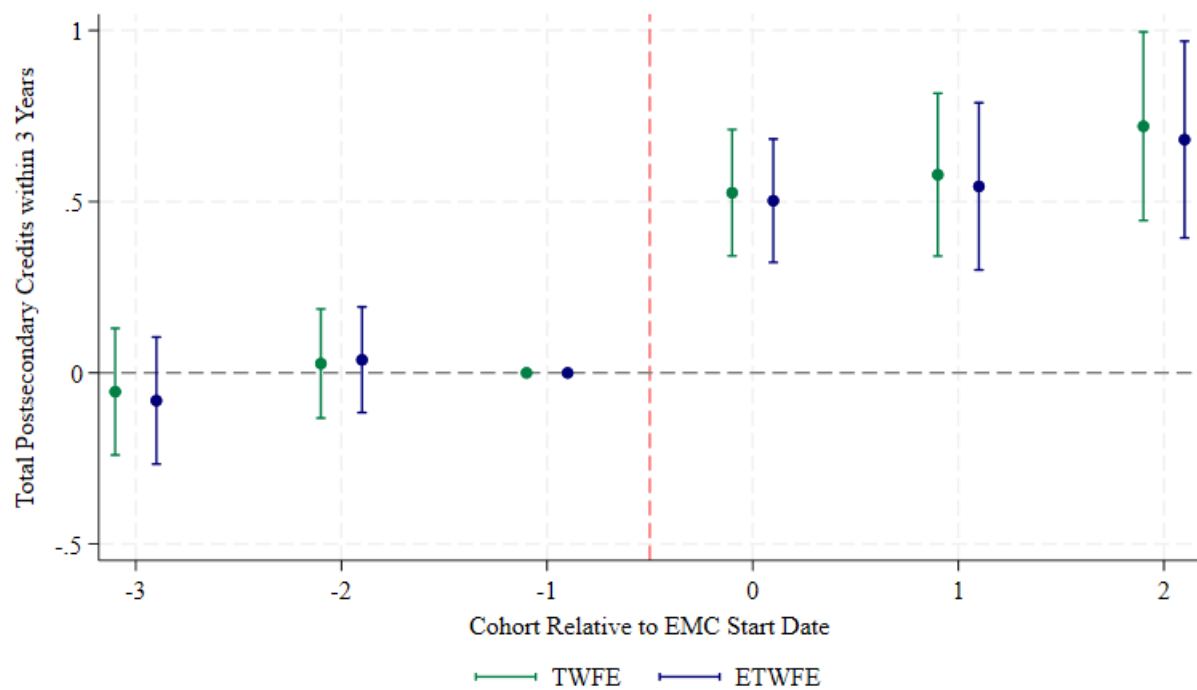
Dashed line indicates the timing of high school graduation on a conventional 4-year schedule.

Figure 5: First Stage Relationship Between EMC Eligibility and Enrollment



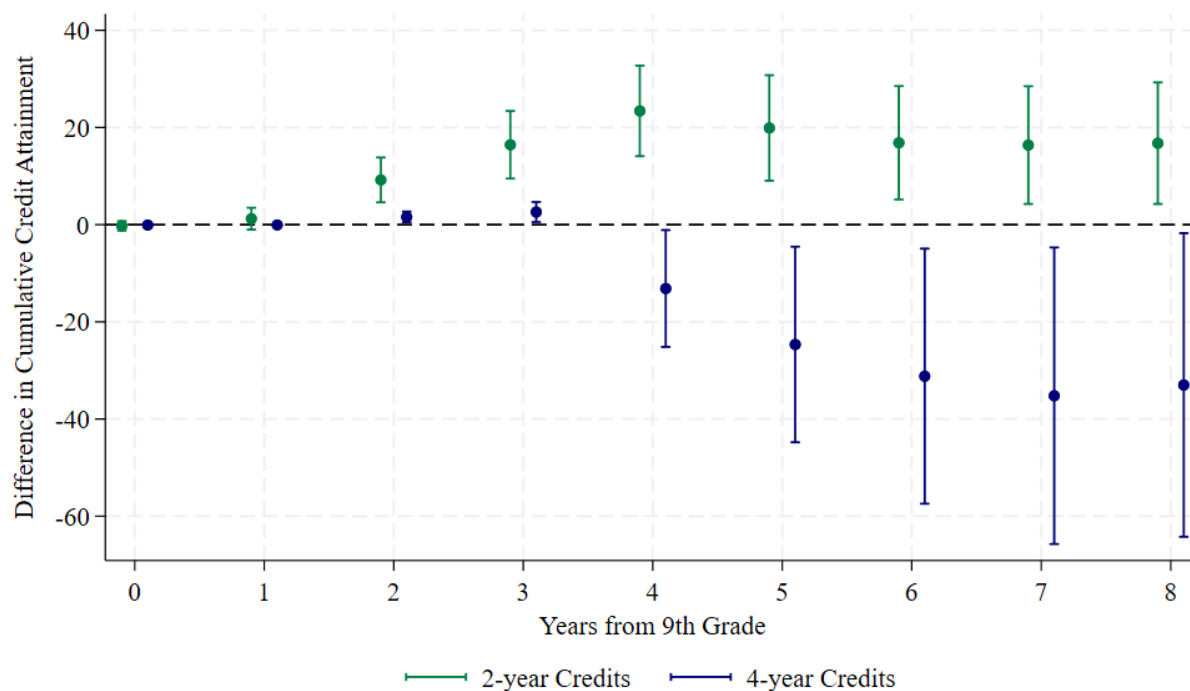
Standard errors are clustered at the school level and bars plot the 95 percent confidence interval.

Figure 6: Reduced Form Relationship Between EMC Eligibility and Credit Attainment between 9th Grade and end of 12th Grade



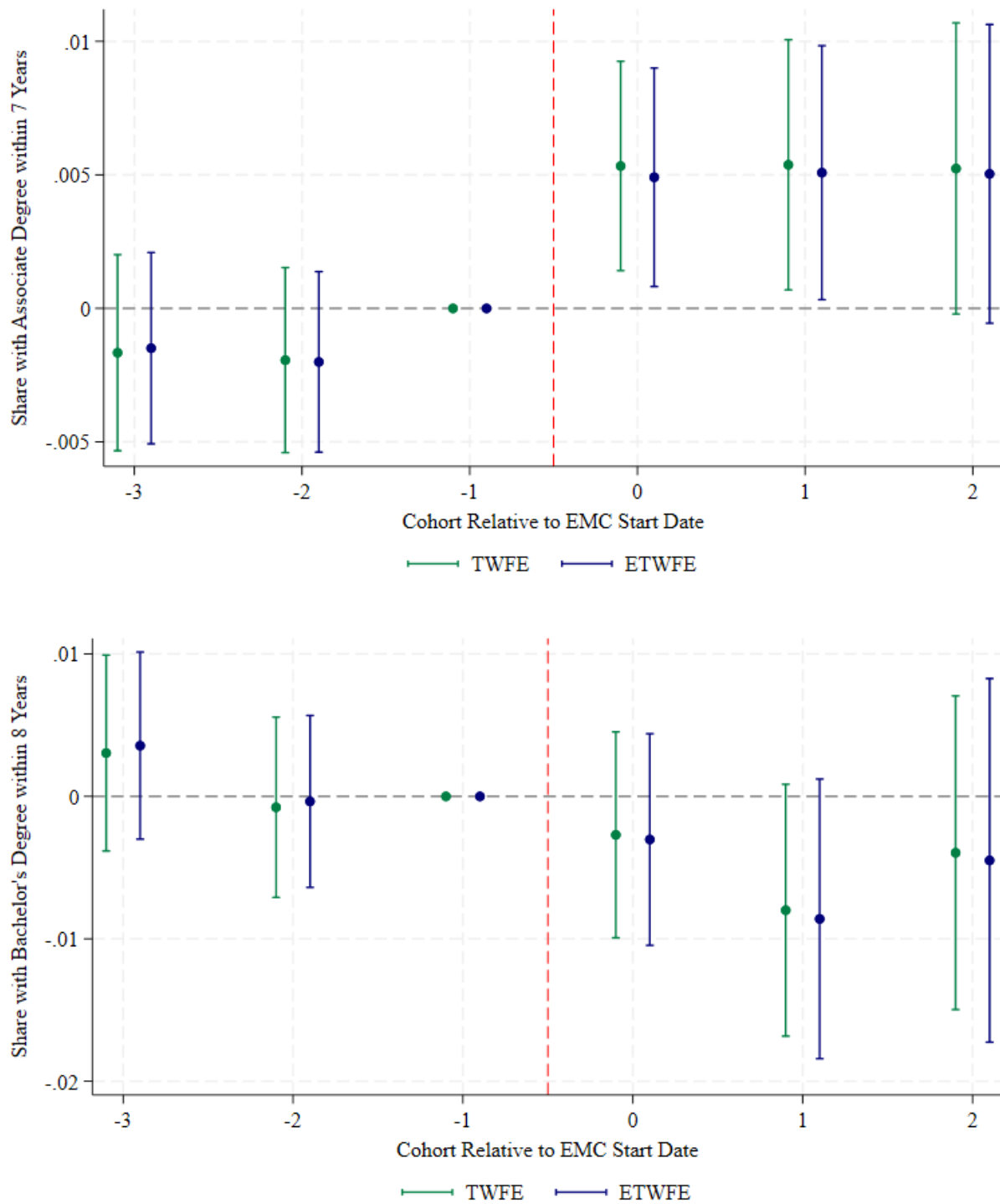
Standard errors are clustered at the school level and bars plot the 95 percent confidence interval.

Figure 7: Two-stage Least Squares Coefficients—Effects of Early Middle College enrollment on credit attainment, by year and credit type



Error bars report a 95 percent confidence interval and standard errors are clustered at the school level. Year 3 is the timing of high school graduation on a standard 4-year schedule. Year 4 is the timing of high school graduation on a standard 5-year EMC schedule. The sample is balanced across time horizons.

Figure 8: Reduced Form Relationship Between EMC Eligibility and Degree Attainment, by IHE Level



Standard errors are clustered at the school level and bars plot the 95 percent confidence interval.

## Appendix A Tables

Table A1: Characteristics by Initial Postsecondary Enrollment

|                       | EMC   | 2-year | 4-yr Priv. | 4-yr Pub. | 4-yr Flagship | No Postsec |
|-----------------------|-------|--------|------------|-----------|---------------|------------|
| SAT Composite Score   | 0.15  | -0.09  | 0.50       | 0.55      | 1.23          | -0.53      |
| 8th Gr. Math Score    | 0.24  | 0.03   | 0.55       | 0.57      | 1.11          | -0.41      |
| 8th Gr. Reading Score | 0.32  | 0.08   | 0.52       | 0.58      | 1.04          | -0.40      |
| Econ. Disadv.         | 0.39  | 0.36   | 0.22       | 0.22      | 0.13          | 0.60       |
| Special Education     | 0.05  | 0.08   | 0.04       | 0.03      | 0.01          | 0.18       |
| ELL                   | 0.02  | 0.04   | 0.01       | 0.02      | 0.01          | 0.03       |
| White                 | 0.79  | 0.76   | 0.81       | 0.80      | 0.75          | 0.72       |
| Black                 | 0.10  | 0.09   | 0.08       | 0.08      | 0.04          | 0.13       |
| Hispanic              | 0.06  | 0.08   | 0.05       | 0.05      | 0.04          | 0.08       |
| Asian                 | 0.01  | 0.02   | 0.03       | 0.04      | 0.13          | 0.01       |
| N                     | 4,309 | 19,887 | 7,295      | 19,913    | 8,512         | 39,339     |

Students who entered high school in EMC eligible cohorts from fall 2014–fall 2018. Column 1 includes EMC enrollees. Column 2 includes students whose initial post-high-school enrollment was at a two-year IHE. Columns 3–5 include students whose initial post-high-school enrollment was at a four-year private, non-flagship public, or flagship IHE, respectively. Column 6 includes students who never enrolled for more than five credits in a semester after high-school graduation.

Table A2: Comparison of Adjacent Cohorts - Pre-Treatment vs. Post-Treatment

|                            | Cohort Before Treatment | Cohort After Treatment |
|----------------------------|-------------------------|------------------------|
| Male                       | 0.509                   | 0.506                  |
| White                      | 0.761                   | 0.747                  |
| Black                      | 0.106                   | 0.107                  |
| Hispanic                   | 0.063                   | 0.071                  |
| Asian                      | 0.030                   | 0.033                  |
| Other                      | 0.040                   | 0.042                  |
| English Language Learner   | 0.025                   | 0.031                  |
| Economically Disadvantaged | 0.390                   | 0.397                  |
| Special Education          | 0.120                   | 0.117                  |
| 8th Grade Math             | 0.146                   | 0.113                  |
| 8th Grade ELA              | 0.104                   | 0.086                  |
| N                          | 31,155                  | 31,731                 |

This table shows the average characteristics of students at treated schools in the cohort that was one year too old to enroll in the EMC and the characteristics of the first cohort that was eligible to enroll.

Table A3: Effects of EMC Enrollment on High School Outcomes

|                                         | Main                            | Balanced ( $t = 0$ )            | Balanced ( $t \leq 2$ )         |
|-----------------------------------------|---------------------------------|---------------------------------|---------------------------------|
| 2-year credits by end of 4th year of HS | 15.938***<br>(2.744)<br>406,037 | 16.441***<br>(3.326)<br>350,263 | 16.908***<br>(3.005)<br>381,357 |
| 4-year credits by end of 4th year of HS | 1.371*<br>(0.812)<br>406,037    | 2.215**<br>(0.961)<br>350,263   | 1.798**<br>(0.882)<br>381,357   |
| Total credits by end of 4th year of HS  | 17.309***<br>(2.805)<br>406,037 | 18.656***<br>(3.377)<br>350,263 | 18.705***<br>(2.988)<br>381,357 |
| 4-year high school graduation           | -0.056<br>(0.096)<br>404,984    | -0.055<br>(0.119)<br>349,413    | -0.067<br>(0.107)<br>380,360    |
| 5-year high school graduation           | -0.024<br>(0.084)<br>405,226    | -0.027<br>(0.098)<br>349,578    | -0.009<br>(0.093)<br>380,589    |
| 6-year high school graduation           | -0.028<br>(0.081)<br>405,235    | -0.003<br>(0.093)<br>349,586    | -0.022<br>(0.089)<br>380,596    |
| SAT Composite                           | -0.320<br>(0.283)<br>232,489    | -0.499<br>(0.333)<br>197,679    | -0.299<br>(0.315)<br>215,827    |

Standard errors are clustered at the school level and listed in parentheses. The Main column reports the baseline specification. Balanced ( $t = 0$ ) restricts the sample to use only  $t = 0$  as the treated period. Balanced ( $t \leq 2$ ) restricts the sample to observations with at least  $t = 2$  available and uses the full  $t = 0, 1, 2$  post-treatment periods. All specifications control for 8th grade math and reading scores, economically disadvantaged status, special education status, English language learner status, race, and sex. Both stages include school and year fixed effects. \*  $p < 0.10$ ; \*\*  $p < 0.05$ ; \*\*\*  $p < 0.01$ .

Table A4: Effects of EMC Enrollment on Postsecondary Attainment

|                                         | Main                             | Balanced ( $t = 0$ )            | Balanced ( $t \leq 2$ )          |
|-----------------------------------------|----------------------------------|---------------------------------|----------------------------------|
| Any postsecondary enrollment in 7 years | 0.090<br>(0.129)<br>369,834      | 0.075<br>(0.153)<br>322,006     | 0.093<br>(0.137)<br>350,785      |
| Any 2-year enrollment in 7 years        | 0.016<br>(0.174)<br>369,834      | -0.038<br>(0.214)<br>322,006    | 0.082<br>(0.183)<br>350,785      |
| Any 4-year enrollment in 7 years        | -0.190<br>(0.116)<br>369,834     | -0.107<br>(0.150)<br>322,006    | -0.212*<br>(0.119)<br>350,785    |
| Any degree in 7 years                   | 0.215***<br>(0.051)<br>369,834   | 0.228***<br>(0.068)<br>322,006  | 0.220***<br>(0.053)<br>350,785   |
| Any degree in 8 years                   | -0.032<br>(0.104)<br>322,525     | 0.048<br>(0.130)<br>290,977     | -0.036<br>(0.105)<br>309,084     |
| Associate degree in 5 years             | 0.204***<br>(0.028)<br>406,037   | 0.183***<br>(0.030)<br>350,263  | 0.216***<br>(0.031)<br>381,357   |
| Associate degree in 6 years             | 0.215***<br>(0.035)<br>406,037   | 0.235***<br>(0.045)<br>350,263  | 0.219***<br>(0.039)<br>381,357   |
| Associate degree in 7 years             | 0.219***<br>(0.048)<br>369,834   | 0.230***<br>(0.063)<br>322,006  | 0.221***<br>(0.050)<br>350,785   |
| Associate degree in 8 years             | 0.158***<br>(0.060)<br>322,525   | 0.235***<br>(0.076)<br>290,977  | 0.156***<br>(0.060)<br>309,084   |
| Bachelor's degree in 8 years            | -0.145<br>(0.098)<br>322,525     | -0.127<br>(0.127)<br>290,977    | -0.146<br>(0.099)<br>309,084     |
| 2-year credits in 7 years               | 15.613***<br>(5.591)<br>369,834  | 19.706***<br>(7.347)<br>322,006 | 17.617***<br>(5.660)<br>350,785  |
| 4-year credits in 7 years               | -35.273**<br>(14.009)<br>369,834 | -30.554*<br>(17.471)<br>322,006 | -37.336**<br>(14.494)<br>350,785 |
| Total credits in 7 years                | -19.660<br>(13.745)<br>369,834   | -10.849<br>(17.292)<br>322,006  | -19.720<br>(14.120)<br>350,785   |

See Table A3 for details and definitions. Outcomes of the form X within T years are defined as follows: 0 years means during 9th grade, 3 years means during the standard 4-year high school enrollment, etc.

Table A5: Effects of EMC Enrollment on Postsecondary Pathway and Transfer Timing

|                                           | Main                            | Balanced ( $t = 0$ )            | Balanced ( $t \leq 2$ )         |
|-------------------------------------------|---------------------------------|---------------------------------|---------------------------------|
| Never reaches 12 credits in either sector | -0.050<br>(0.079)<br>270,756    | -0.122<br>(0.094)<br>233,414    | -0.094<br>(0.089)<br>254,468    |
| Only reaches 12 credits in 2-yr sector    | 0.228***<br>(0.078)<br>270,756  | 0.249***<br>(0.094)<br>233,414  | 0.269***<br>(0.086)<br>254,468  |
| Only reaches 12 credits in 4-yr sector    | -0.235**<br>(0.111)<br>270,756  | -0.239*<br>(0.137)<br>233,414   | -0.244**<br>(0.120)<br>254,468  |
| Reaches 12 credits in 2-yr then 4-yr      | 0.110<br>(0.076)<br>270,756     | 0.174*<br>(0.093)<br>233,414    | 0.121<br>(0.083)<br>254,468     |
| Reaches 12 credits in 4-yr then 2-yr      | -0.045<br>(0.032)<br>270,756    | -0.057<br>(0.048)<br>233,414    | -0.044<br>(0.034)<br>254,468    |
| Years to 12 2-yr credits                  | -1.353***<br>(0.247)<br>98,440  | -1.502***<br>(0.285)<br>83,848  | -1.346***<br>(0.283)<br>91,994  |
| Years to 12 4-yr credits                  | 0.026<br>(0.310)<br>164,995     | -0.229<br>(0.437)<br>141,990    | 0.041<br>(0.345)<br>154,929     |
| Years to 12 postsecondary credits         | -1.117***<br>(0.221)<br>224,084 | -1.321***<br>(0.265)<br>192,459 | -1.120***<br>(0.248)<br>210,253 |
| Years between 12 2-yr and 12 4-yr credits | 0.942***<br>(0.246)<br>29,062   | 0.919**<br>(0.364)<br>24,341    | 0.998***<br>(0.276)<br>26,968   |

See Table A3 for details and definitions. A sixth pathway—contemporaneously reaching 12 2-year and 12 4-year credits—was estimated but is omitted for brevity because the effect was small and statistically insignificant.

Table A6: Effects of EMC Enrollment on High School Outcomes

| Outcome                                 | Main Estimates |         |         | Admin Treatment Date |         |         |
|-----------------------------------------|----------------|---------|---------|----------------------|---------|---------|
|                                         | Coefficient    | SE      | N       | Coefficient          | SE      | N       |
| 2-year credits by end of 4th year of HS | 15.938***      | (2.744) | 406,037 | 21.703***            | (5.893) | 370,813 |
| 4-year credits by end of 4th year of HS | 1.371*         | (0.812) | 406,037 | -8.113               | (9.132) | 370,813 |
| Total credits by end of 4th year of HS  | 17.309***      | (2.805) | 406,037 | 13.589               | (9.398) | 370,813 |
| 4-year high school graduation           | -0.056         | (0.096) | 404,984 | 0.064                | (0.164) | 369,851 |
| 5-year high school graduation           | -0.024         | (0.084) | 405,226 | 0.040                | (0.143) | 370,072 |
| 6-year high school graduation           | -0.028         | (0.081) | 405,235 | 0.008                | (0.134) | 370,077 |
| SAT Composite                           | -0.320         | (0.283) | 232,489 | -0.433               | (0.500) | 212,197 |

Standard errors are clustered at the school level. Estimates are based on a 2SLS specification controlling for 8th grade math and reading scores, economically disadvantaged status, special education status, English language learner status, race, and sex. Both the first and second stage regressions include school and year fixed effects. Outcomes of the form X within T years are defined as follows: 0 years means during 9th grade, 3 years means during the standard 4-year high school enrollment, etc. \* p<0.10; \*\* p<0.05; \*\*\* p<0.01.

Table A7: Effects of EMC Enrollment on Postsecondary Outcomes

| Outcome                               | Main Sample |          |         | Admin Treatment Date |          |         |
|---------------------------------------|-------------|----------|---------|----------------------|----------|---------|
|                                       | Coefficient | SE       | N       | Coefficient          | SE       | N       |
| Any postsec enrollment within 7 years | 0.090       | (0.129)  | 369,834 | 0.104                | (0.234)  | 370,813 |
| Any 2-year enrollment within 7 years  | 0.016       | (0.174)  | 369,834 | 0.003                | (0.314)  | 370,813 |
| Any 4-year enrollment within 7 years  | -0.190      | (0.116)  | 369,834 | -0.046               | (0.196)  | 370,813 |
| Any degree within 7 years             | 0.215***    | (0.051)  | 369,834 | 0.178**              | (0.088)  | 343,045 |
| Any degree within 8 years             | -0.032      | (0.104)  | 322,525 | 0.069                | (0.191)  | 300,826 |
| Associate degree within 5 years       | 0.204***    | (0.028)  | 406,037 | 0.188***             | (0.039)  | 370,813 |
| Associate degree within 6 years       | 0.215***    | (0.035)  | 406,037 | 0.183***             | (0.058)  | 370,813 |
| Associate degree within 7 years       | 0.219***    | (0.048)  | 369,834 | 0.201***             | (0.074)  | 343,045 |
| Associate degree within 8 years       | 0.158***    | (0.060)  | 322,525 | 0.209*               | (0.107)  | 300,826 |
| Bachelor's degree within 8 years      | -0.145      | (0.098)  | 322,525 | -0.110               | (0.180)  | 300,826 |
| 2-year credits within 7 years         | 15.613***   | (5.591)  | 369,834 | 17.167*              | (9.553)  | 343,045 |
| 4-year credits within 7 years         | -35.273**   | (14.009) | 369,834 | -15.274              | (26.599) | 343,045 |
| Total credits within 7 years          | -19.660     | (13.745) | 369,834 | 1.893                | (25.403) | 343,045 |

Standard errors are clustered at the school level. Estimates are based on a 2SLS specification controlling for 8th grade math and reading scores, economically disadvantaged status, special education status, English language learner status, race, and sex. Both the first and second stage regressions include school and year fixed effects. Outcomes of the form X within T years are defined as follows: 0 years means during 9th grade, 3 years means during the standard 4-year high school enrollment, etc. \* p<0.10; \*\* p<0.05; \*\*\* p<0.01.

Table A8: Effects of EMC Enrollment on Postsecondary Pathway and Transfer Timing

| Outcome                                       | Main Sample |         |         | Admin Treatment Date |         |         |
|-----------------------------------------------|-------------|---------|---------|----------------------|---------|---------|
|                                               | Coefficient | SE      | N       | Coefficient          | SE      | N       |
| Never reached 12 postsec credits              | -0.050      | (0.079) | 270,756 | -0.309**             | (0.146) | 249,400 |
| Only reached 12 2-year credits                | 0.228***    | (0.078) | 270,756 | 0.378***             | (0.133) | 249,400 |
| Only reached 12 4-year credits                | -0.235**    | (0.111) | 270,756 | -0.205               | (0.196) | 249,400 |
| Reached 12 2-year, then 12 4-year credits     | 0.110       | (0.076) | 270,756 | 0.097                | (0.131) | 249,400 |
| Reached 12 4-year, then 12 2-year credits     | -0.045      | (0.032) | 270,756 | 0.051                | (0.058) | 249,400 |
| Years to 12 2-year credits                    | -1.353***   | (0.247) | 98,440  | -1.396***            | (0.366) | 89,480  |
| Years to 12 4-year credits                    | 0.026       | (0.310) | 164,995 | -0.526               | (0.521) | 153,216 |
| Years to 12 total credits                     | -1.117***   | (0.221) | 224,084 | -1.347***            | (0.328) | 206,539 |
| Years between 12 2-year and 12 4-year credits | 0.942***    | (0.245) | 29,062  | 0.539                | (0.374) | 26,528  |

Standard errors are clustered at the school level. Estimates are based on a 2SLS specification controlling for 8th grade math and reading scores, economically disadvantaged status, special education status, English language learner status, race, and sex. Both the first and second stage regressions include school and year fixed effects. Outcomes of the form X within T years are defined as follows: 0 years means during 9th grade, 3 years means during the standard 4-year high school enrollment, etc. \* p<0.10; \*\* p<0.05; \*\*\* p<0.01.

Table A9: Effects of EMC Enrollment on Degree Attainment: Common Sample

| Outcome                   | Main Sample |         |         | Common Sample |         |         |
|---------------------------|-------------|---------|---------|---------------|---------|---------|
|                           | Coefficient | SE      | N       | Coefficient   | SE      | N       |
| Any degree within 7 years | 0.215***    | (0.051) | 369,834 | 0.173***      | (0.059) | 322,525 |
| Any degree within 8 years | -0.032      | (0.104) | 322,525 | -0.032        | (0.104) | 322,525 |

Standard errors are clustered at the school level. Estimates are based on a 2SLS specification controlling for 8th grade math and reading scores, economically disadvantaged status, special education status, English language learner status, race, and sex. Both the first and second stage regressions include school and year fixed effects. The common-sample estimates restrict the sample to observations with nonmissing outcomes for all comparable time horizons. Outcomes of the form X within T years are defined as follows: 0 years means during 9th grade, 3 years means during the standard 4-year high school enrollment, etc. \* p<0.10; \*\* p<0.05; \*\*\* p<0.01.

Table A10: Effects of EMC Enrollment on Associate Degree Attainment: Common Sample

| Outcome                         | Main Sample |         |         | Common Sample |         |         |
|---------------------------------|-------------|---------|---------|---------------|---------|---------|
|                                 | Coefficient | SE      | N       | Coefficient   | SE      | N       |
| Associate degree within 5 years | 0.204***    | (0.028) | 406,037 | 0.204***      | (0.032) | 322,525 |
| Associate degree within 6 years | 0.215***    | (0.035) | 406,037 | 0.202***      | (0.044) | 322,525 |
| Associate degree within 7 years | 0.219***    | (0.048) | 369,834 | 0.178***      | (0.055) | 322,525 |
| Associate degree within 8 years | 0.158***    | (0.060) | 322,525 | 0.158***      | (0.060) | 322,525 |

Standard errors are clustered at the school level. Estimates are based on a 2SLS specification controlling for 8th grade math and reading scores, economically disadvantaged status, special education status, English language learner status, race, and sex. Both the first and second stage regressions include school and year fixed effects. The common-sample estimates restrict the sample to observations with nonmissing outcomes for all comparable time horizons. Outcomes of the form X within T years are defined as follows: 0 years means during 9th grade, 3 years means during the standard 4-year high school enrollment, etc. \* p<0.10; \*\* p<0.05; \*\*\* p<0.01.

Table A11: Effects of EMC Enrollment on High School Outcomes: No Never-Treated Comparison

| Outcome                                 | Main Estimates |         |         | No Never-Treated |         |         |
|-----------------------------------------|----------------|---------|---------|------------------|---------|---------|
|                                         | Coefficient    | SE      | N       | Coefficient      | SE      | N       |
| 2-year credits by end of 4th year of HS | 15.938***      | (2.744) | 406,037 | 20.984***        | (4.353) | 182,379 |
| 4-year credits by end of 4th year of HS | 1.371*         | (0.812) | 406,037 | -4.052           | (4.804) | 182,379 |
| Total credits by end of 4th year of HS  | 17.309***      | (2.805) | 406,037 | 16.932***        | (5.848) | 182,379 |
| 4-year high school graduation           | -0.056         | (0.096) | 404,984 | -0.154           | (0.137) | 181,824 |
| 5-year high school graduation           | -0.024         | (0.084) | 405,226 | -0.152           | (0.098) | 181,998 |
| 6-year high school graduation           | -0.028         | (0.081) | 405,235 | -0.137           | (0.091) | 182,004 |
| SAT Composite                           | -0.320         | (0.283) | 232,489 | 0.143            | (0.382) | 118,582 |

Standard errors are clustered at the school level. Estimates are based on a 2SLS specification controlling for 8th grade math and reading scores, economically disadvantaged status, special education status, English language learner status, race, and sex. Both the first and second stage regressions include school and year fixed effects. The no-never-treated estimates exclude observations from schools that never implement EMC. Outcomes of the form X within T years are defined as follows: 0 years means during 9th grade, 3 years means during the standard 4-year high school enrollment, etc. \*  $p < 0.10$ ; \*\*  $p < 0.05$ ; \*\*\*  $p < 0.01$ .

Table A12: Effects of EMC Enrollment on Postsecondary Outcomes: No Never-Treated Comparison

| Outcome                               | Main Estimates |          |         | No Never-Treated |          |         |
|---------------------------------------|----------------|----------|---------|------------------|----------|---------|
|                                       | Coefficient    | SE       | N       | Coefficient      | SE       | N       |
| Any postsec enrollment within 7 years | 0.090          | (0.129)  | 369,834 | 0.148            | (0.165)  | 182,379 |
| Any 2-year enrollment within 7 years  | 0.016          | (0.174)  | 369,834 | -0.014           | (0.170)  | 182,379 |
| Any 4-year enrollment within 7 years  | -0.190         | (0.116)  | 369,834 | -0.114           | (0.146)  | 182,379 |
| Any degree within 7 years             | 0.215***       | (0.051)  | 369,834 | 0.223***         | (0.070)  | 171,732 |
| Any degree within 8 years             | -0.032         | (0.104)  | 322,525 | 0.026            | (0.148)  | 149,844 |
| Associate degree within 5 years       | 0.204***       | (0.028)  | 406,037 | 0.183***         | (0.033)  | 182,379 |
| Associate degree within 6 years       | 0.215***       | (0.035)  | 406,037 | 0.215***         | (0.044)  | 182,379 |
| Associate degree within 7 years       | 0.219***       | (0.048)  | 369,834 | 0.224***         | (0.067)  | 171,732 |
| Associate degree within 8 years       | 0.158***       | (0.060)  | 322,525 | 0.106            | (0.090)  | 149,844 |
| Bachelor's degree within 8 years      | -0.145         | (0.098)  | 322,525 | -0.048           | (0.143)  | 149,844 |
| 2-year credits within 7 years         | 15.613***      | (5.591)  | 369,834 | 18.050**         | (7.866)  | 171,732 |
| 4-year credits within 7 years         | -35.273**      | (14.009) | 369,834 | -16.502          | (15.237) | 171,732 |
| Total credits within 7 years          | -19.660        | (13.745) | 369,834 | 1.548            | (16.019) | 171,732 |

Standard errors are clustered at the school level. Estimates are based on a 2SLS specification controlling for 8th grade math and reading scores, economically disadvantaged status, special education status, English language learner status, race, and sex. Both the first and second stage regressions include school and year fixed effects. The no-never-treated estimates exclude observations from schools that never implement EMC. Outcomes of the form X within T years are defined as follows: 0 years means during 9th grade, 3 years means during the standard 4-year high school enrollment, etc. \*  $p < 0.10$ ; \*\*  $p < 0.05$ ; \*\*\*  $p < 0.01$ .

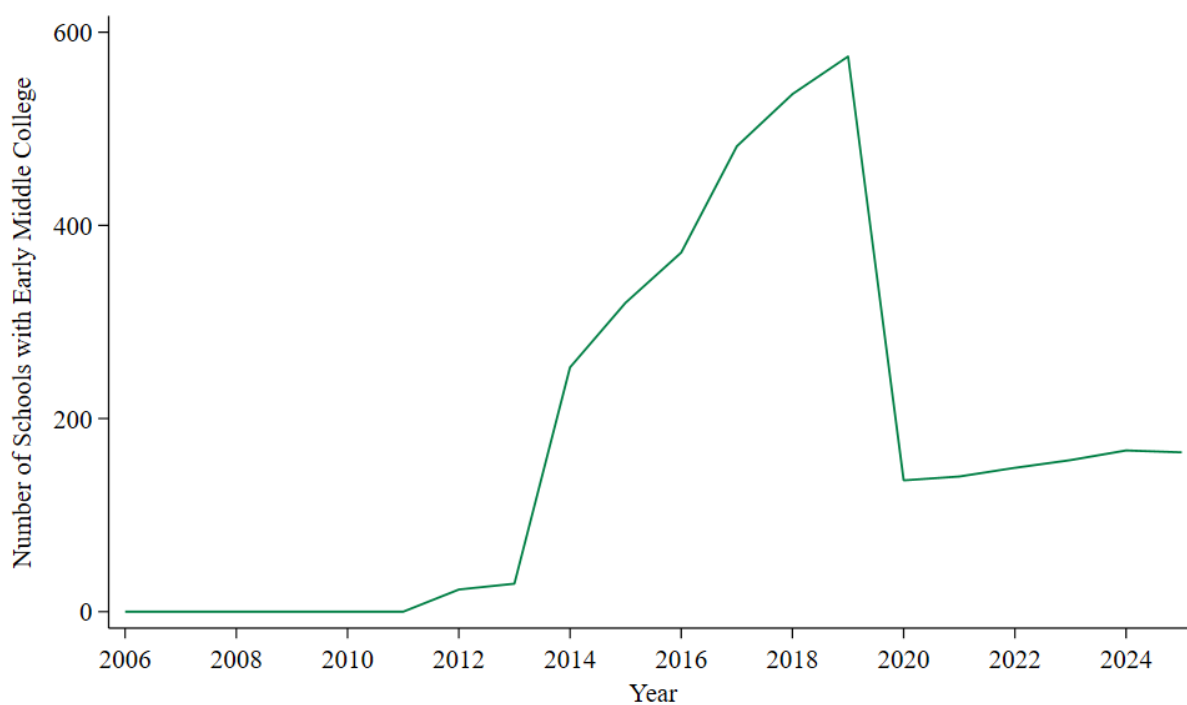
Table A13: Effects of EMC Enrollment on Postsecondary Pathway and Transfer Timing: No Never-Treated Comparison

| Outcome                                       | Main Estimates |         |         | No Never-Treated |         |         |
|-----------------------------------------------|----------------|---------|---------|------------------|---------|---------|
|                                               | Coefficient    | SE      | N       | Coefficient      | SE      | N       |
| Never reached 12 postsec credits              | -0.050         | (0.079) | 270,756 | -0.082           | (0.094) | 126,379 |
| Only reached 12 2-year credits                | 0.228***       | (0.078) | 270,756 | 0.194*           | (0.103) | 126,379 |
| Only reached 12 4-year credits                | -0.235**       | (0.111) | 270,756 | -0.100           | (0.144) | 126,379 |
| Reached 12 2-year, then 12 4-year credits     | 0.110          | (0.076) | 270,756 | 0.114            | (0.094) | 126,379 |
| Reached 12 4-year, then 12 2-year credits     | -0.045         | (0.032) | 270,756 | -0.115**         | (0.051) | 126,379 |
| Years to 12 2-year credits                    | -1.353***      | (0.247) | 98,440  | -1.614***        | (0.288) | 49,750  |
| Years to 12 4-year credits                    | 0.026          | (0.310) | 164,995 | 0.536            | (0.517) | 78,984  |
| Years to 12 total credits                     | -1.117***      | (0.221) | 224,084 | -0.943***        | (0.278) | 108,448 |
| Years between 12 2-year and 12 4-year credits | 0.942***       | (0.245) | 29,062  | 1.484***         | (0.409) | 15,393  |

Standard errors are clustered at the school level. Estimates are based on a 2SLS specification controlling for 8th grade math and reading scores, economically disadvantaged status, special education status, English language learner status, race, and sex. Both the first and second stage regressions include school and year fixed effects. The no-never-treated estimates exclude observations from schools that never implement EMC. Outcomes of the form X within T years are defined as follows: 0 years means during 9th grade, 3 years means during the standard 4-year high school enrollment, etc. \* p<0.10; \*\* p<0.05; \*\*\* p<0.01.

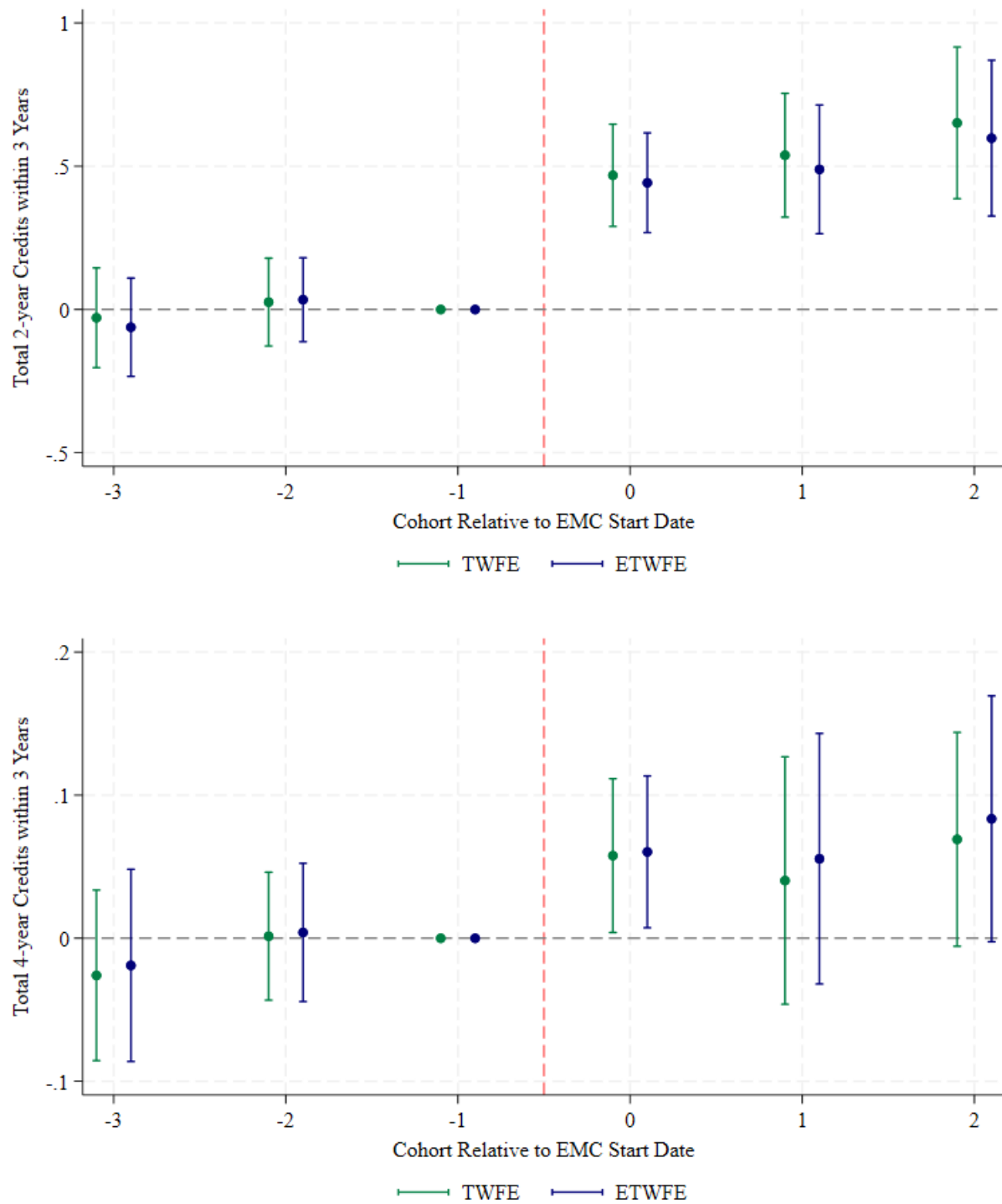
## Appendix B   Figures

Figure B1: Number of Active Early Middle Colleges in Michigan



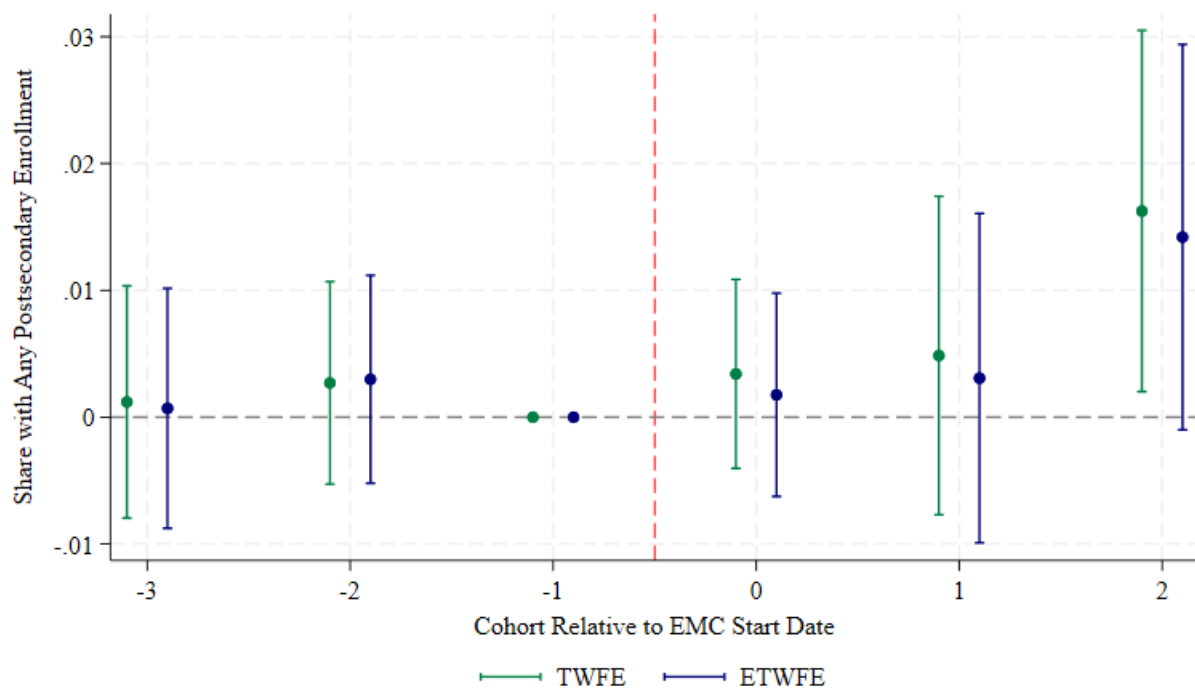
This figure plots the number of active Early Middle College programs and schools reported in the Michigan Educational Entity Master (EEM).

Figure B2: Reduced Form Relationship Between EMC Eligibility and Credit Attainment between 9th and end of 12th Grade, by IHE Type



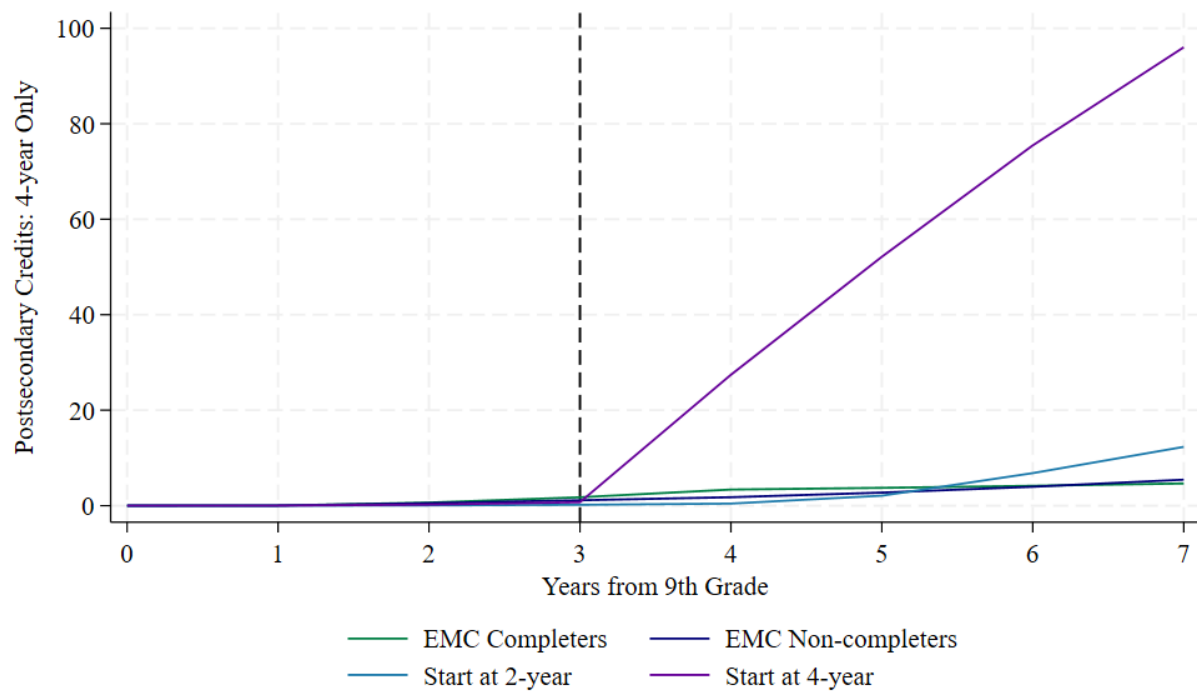
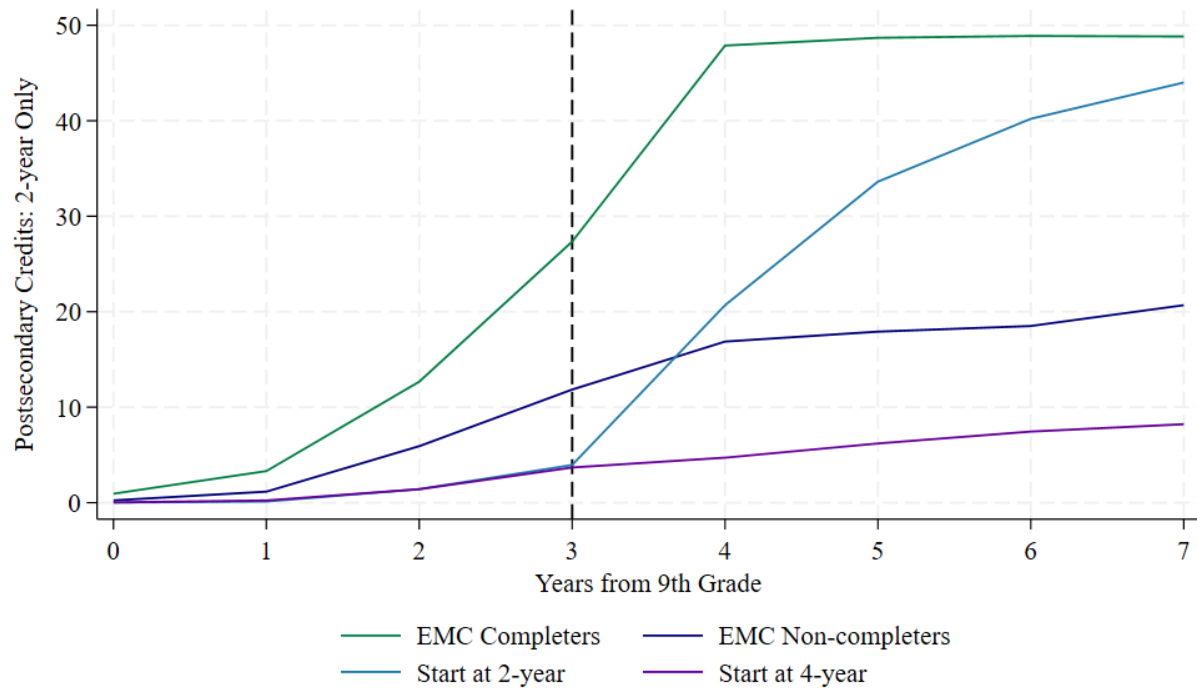
Standard errors are clustered at the school level and bars plot the 95 percent confidence interval.

Figure B3: Reduced Form Relationship Between EMC Eligibility and Ever Enrolling in College



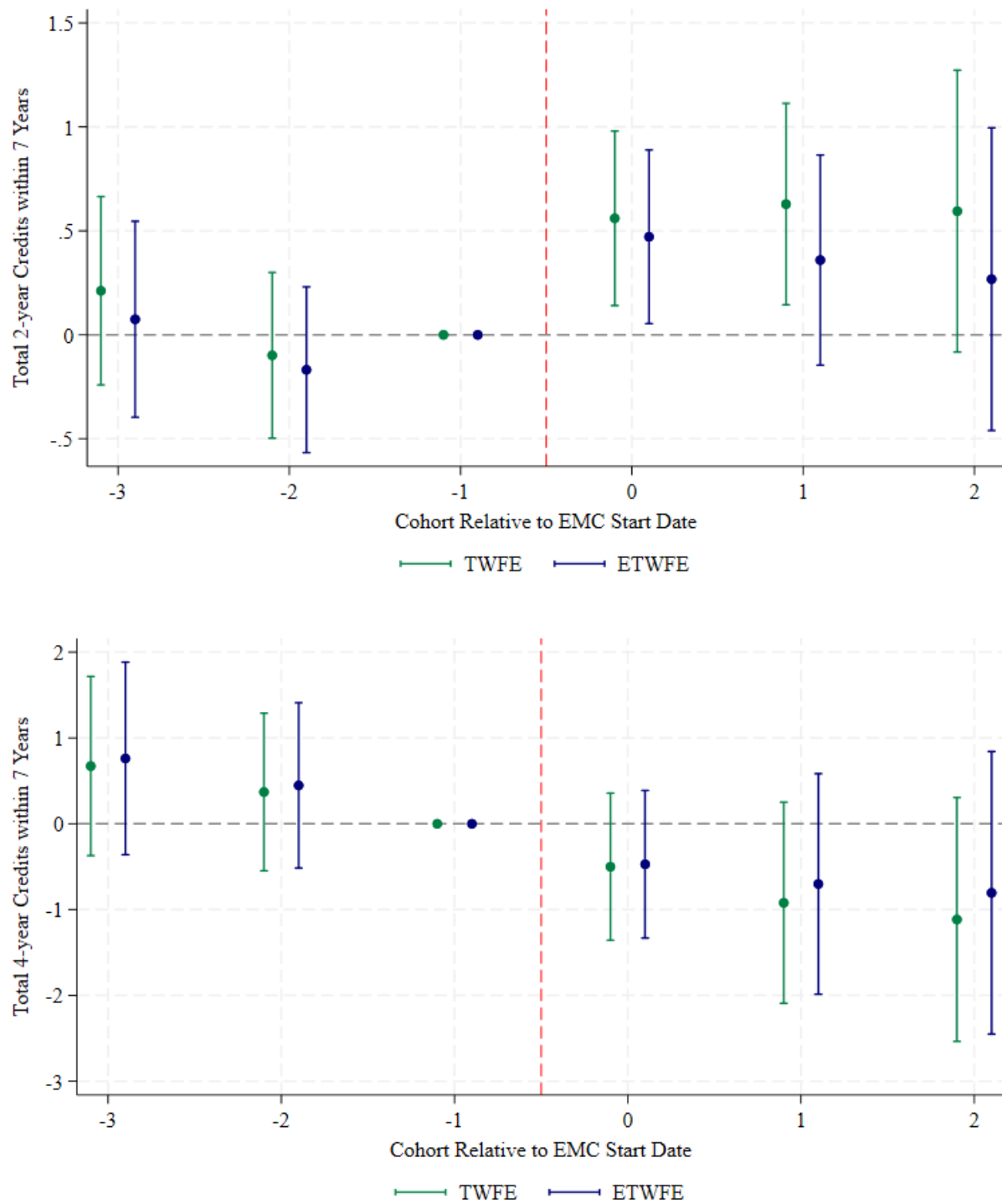
Standard errors are clustered at the school level and bars plot the 95 percent confidence interval.

Figure B4: Average Total Postsecondary Credit Attainment, by Postsecondary Pathway and Credit Level



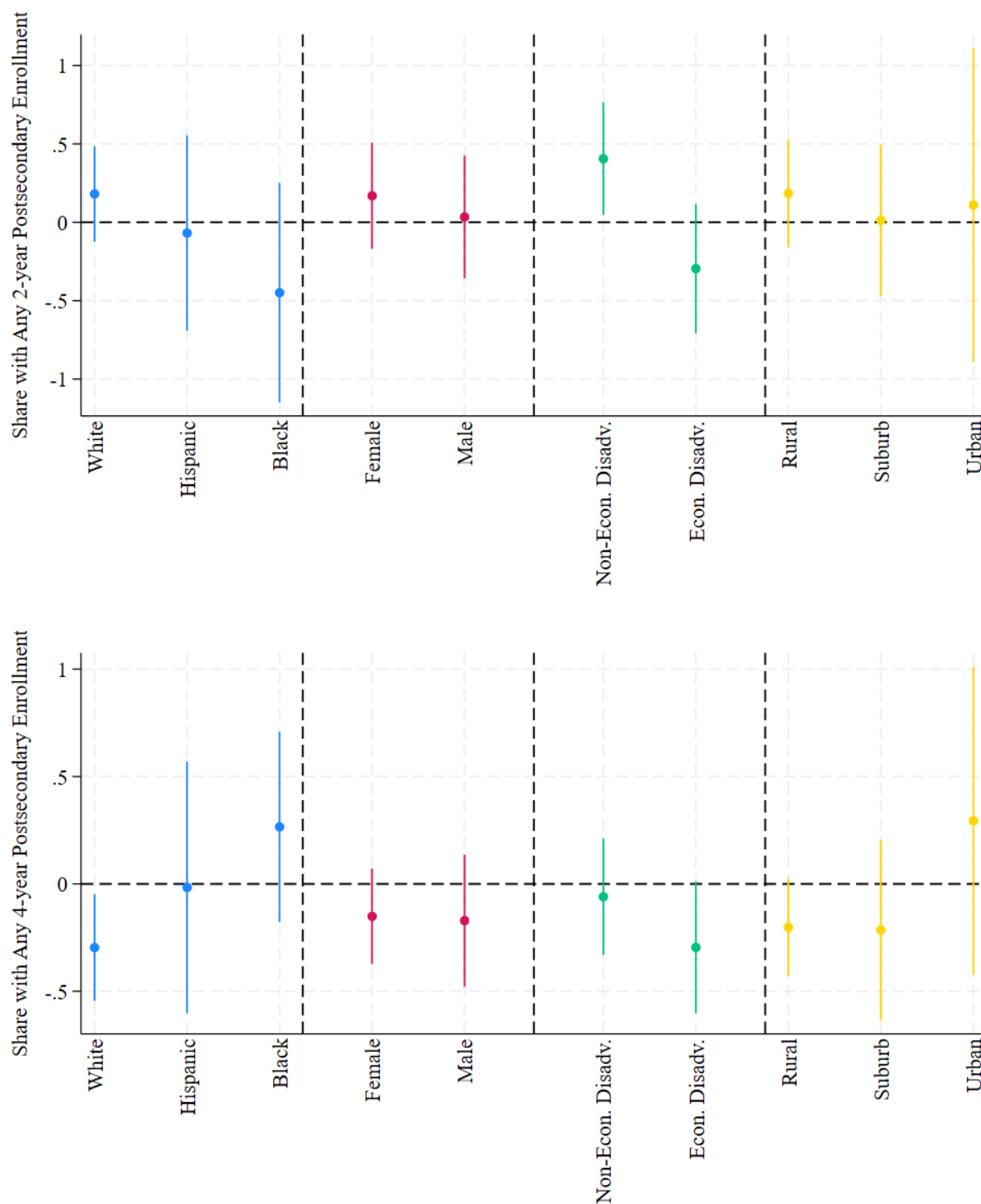
Dashed line indicates the timing of high school graduation on a conventional 4-year schedule.

Figure B5: Reduced Form Relationship Between EMC Eligibility and Credit Attainment within 7 Years of 9th Grade Entry, by IHE Type



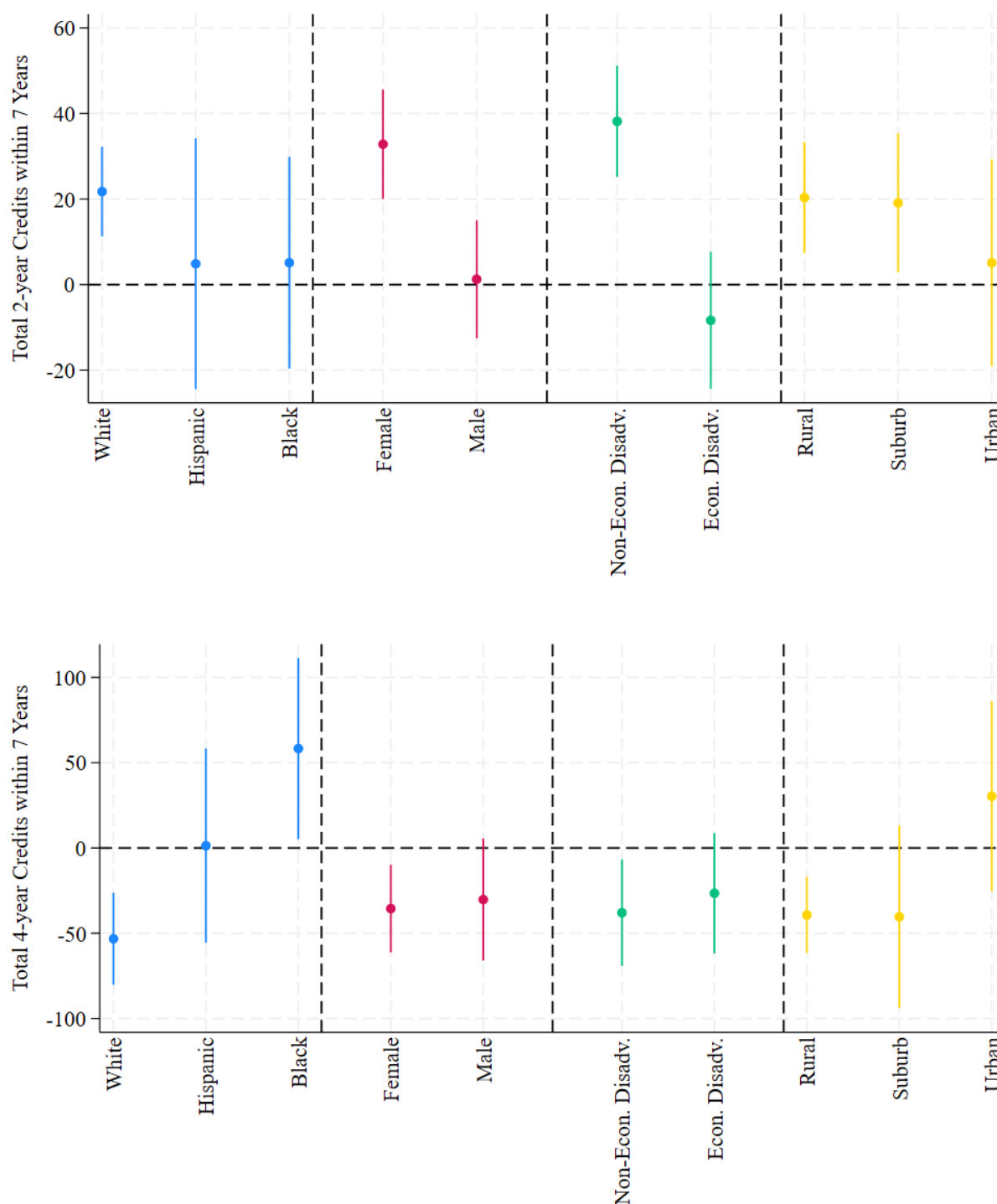
Standard errors are clustered at the school level and bars plot the 95 percent confidence interval.

Figure B6: 2SLS Estimates—Effect of EMC Enrollment on any Post-secondary Enrollment within 7 Years from 9th Grade, by IHE Type and Demographic Group



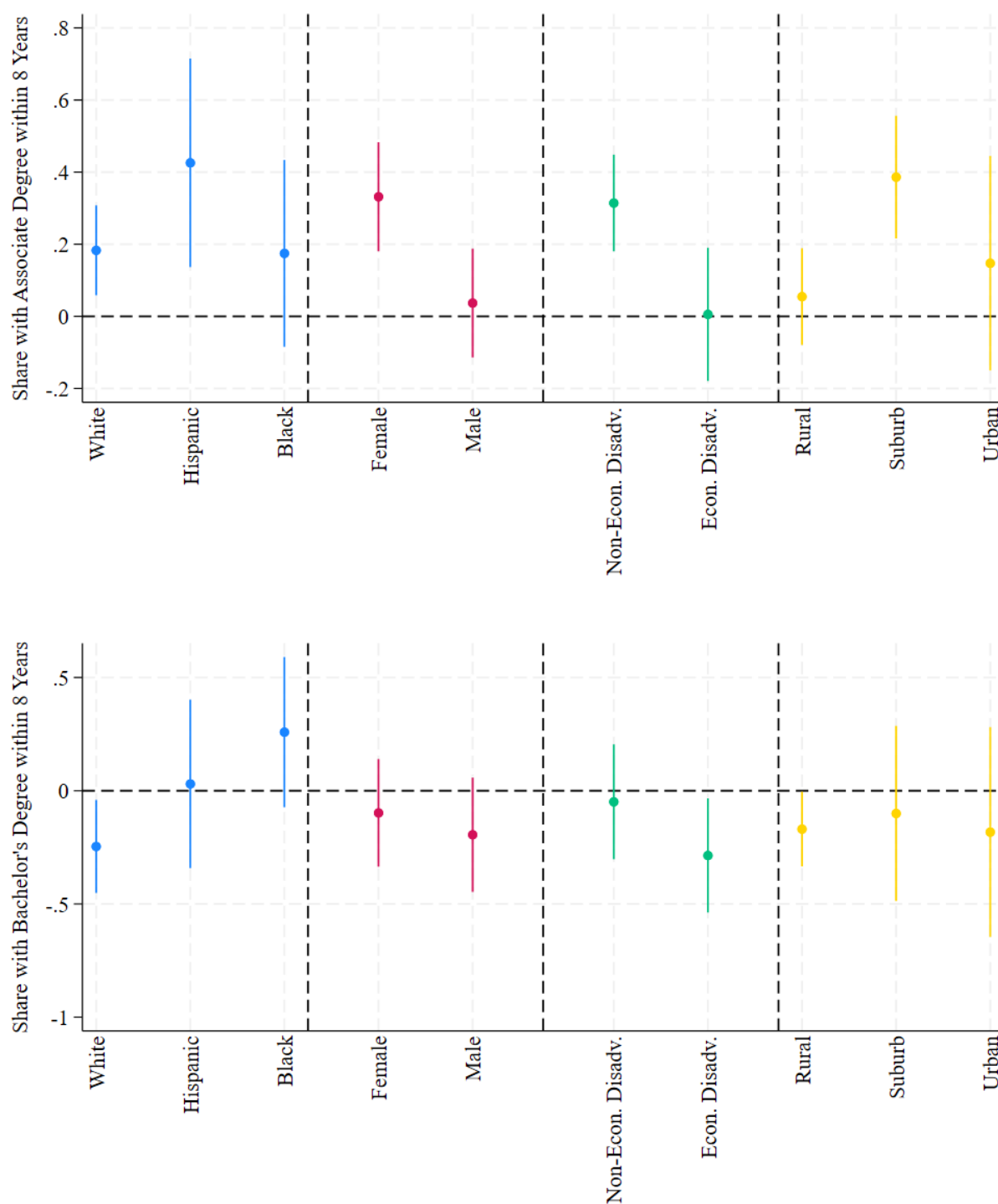
Standard errors are clustered at the school level and bars plot the 95 percent confidence interval. Estimates are based on a 2SLS specification controlling for 8th grade math and reading scores. Both stages include school and year fixed effects. Outcomes of the form X within T years are defined as follows: 0 years means during 9th grade, etc.

Figure B7: 2SLS Estimates–Effect of EMC Enrollment on Post-secondary Credits within 7 Years from 9th Grade, by IHE Type and Demographic Group



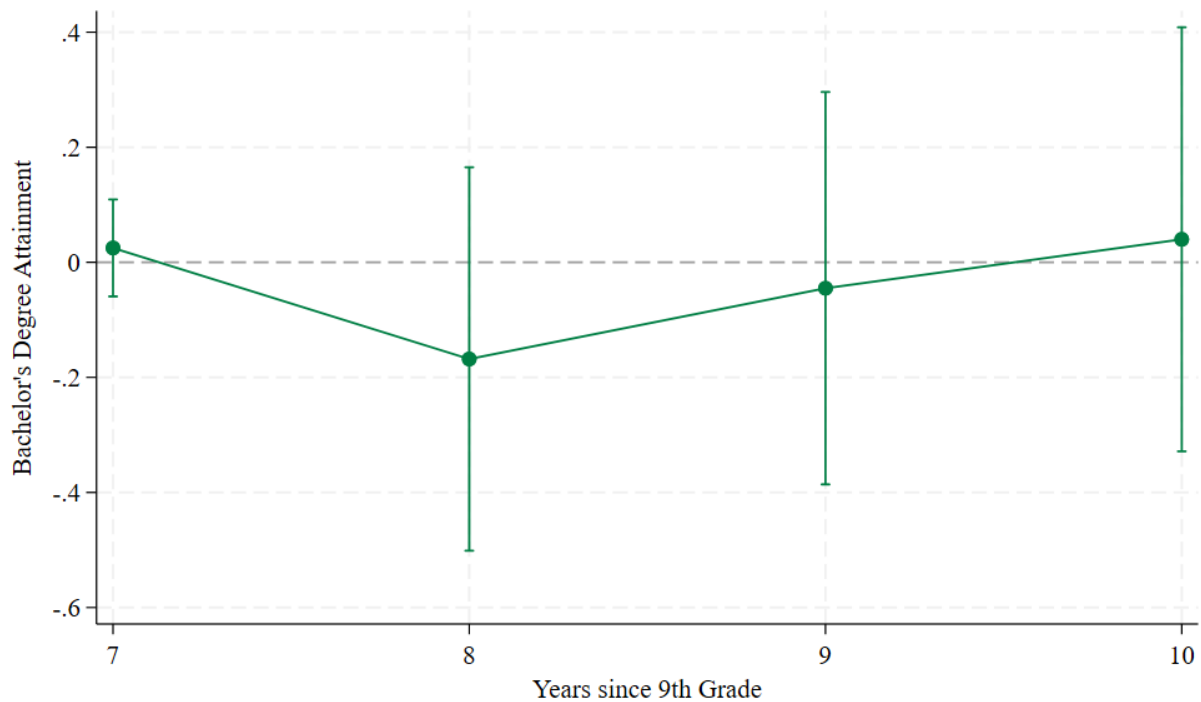
Standard errors are clustered at the school level and bars plot the 95 percent confidence interval. Estimates are based on a 2SLS specification controlling for 8th grade math and reading scores. Both stages include school and year fixed effects. Outcomes of the form X within T years are defined as follows: 0 years means during 9th grade, etc.

Figure B8: 2SLS Estimates—Effect of EMC Enrollment on Degree Attainment within 8 Years from 9th Grade, by IHE Type and Demographic Group



Standard errors are clustered at the school level and bars plot the 95 percent confidence interval. Estimates are based on a 2SLS specification controlling for 8th grade math and reading scores. Both stages include school and year fixed effects. Outcomes of the form X within T years are defined as follows: 0 years means during 9th grade, etc.

Figure B9: 2SLS Estimates—Effect of EMC Enrollment on Bachelor’s Degree Attainment, Fall 2011 through Fall 2014 9th Graders



This analysis uses a restricted sample of fall 2011 through fall 2014 9th grade cohorts to allow for a balanced sample up to 10 years after 9th grade (7 years after standard high school graduation). Therefore, there is only one cohort of treated students (fall 2014) at treated schools. Standard errors are clustered at the school level and bars plot the 95 percent confidence interval. Estimates are based on a 2SLS specification controlling for 8th grade math and reading scores. Both stages include school and year fixed effects. Outcomes of the form X within T years are defined as follows: 0 years means during 9th grade, etc.